Highlights

Process Mining and Stochastic Modeling for Software Process Forecasting: A Systematic Mapping Study with a Human-Checked LLM-Assisted Screen

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- Pre-registered human check of an LLM screen: Recall 0.630 and 0.725 vs a 0.90 bar.
- Three failure mechanisms of the screen are named and quantified.
- 456-record corpus (380 family-marked) at the PM–stochastic–forecasting intersection.
- 99 two-family studies re-examined; none reaches L3 (PRIMAD remains L2).
- Replication package: Zenodo DOI 10.5281/zenodo.21939600.

Process Mining and Stochastic Modeling for Software Process Forecasting: A Systematic Mapping Study with a Human-Checked LLM-Assisted Screen

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Abstract

Context. PATHCAST proposes a forecasting pipeline that couples process mining of repository event logs, a stochastic model of the mined process, Monte Carlo forecasting of lifecycle metrics, and machine-learning correction, with explicit interfaces validated end-to-end. Before building it, we asked whether such a pipeline had already been published. **Objective.** To map primary studies combining process mining, stochastic modelling, and forecasting on software development data; to position PATHCAST against them; and to measure the selection error of the LLM-assisted screen. **Method.** A systematic mapping study reported against SEGRESS [1]. Searches of five databases on 12 April 2026 yielded 5,783 deduplicated records. An LLM screened them; two human raters checked pre-registered samples against a 0.90 Recall bar; two reviewers then accepted or rejected suggested labels on the 4,005 excluded or unclassified records (92% acceptance). Studies were classified into four technique and data families and into integration levels L0–L3, where L3 is any pipeline coupling the three families through explicit interfaces validated end-to-end, the level PATHCAST targets. **Results.** The corpus has 456 records, 380 with a family mark and 76 known only by title. On the 151 fully coded working-set studies, 104 use process mining, 49 a stochastic model, and 27 forecasting; eight combine the first two. Among the 380, one (PRIMAD) combines all three, at L2. Every two-family study (n = 99) was re-examined from PDF or abstract for the third; none reaches L3. The screen reached Recall 0.630 (29 of 46) and 0.725 (235 of 324), below the bar; the re-screen added 150 studies. **Conclusion.** Within the reviewed corpus, no published work reports an L3 pipeline, the level PATHCAST targets; the closest, PRIMAD, stops at component-wise validation. Because the screen missed eligible studies, this is a statement about the corpus, not a proof of absence in the literature. The SPMF taxonomy and a four-direction agenda frame the remaining work.

Keywords: Systematic Mapping Study, Process Mining, Stochastic Modeling, Software Process Forecasting, LLM-Assisted Screening, Mining Software Repositories

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1. Introduction

Software organisations routinely need to answer forecasting questions about their own development process: when a feature will be delivered, how long an issue will remain open, or whether a release will slip. The data needed to answer them already exists as a by-product of everyday work—commits, issue transitions, pull requests, code reviews, and CI/CD executions are all time-stamped events. Three research communities have developed techniques that can exploit these traces, largely independently of one another. *Process mining* (PM) discovers, checks, and enhances process models from event logs, and has been applied to software repositories since the work of Cook and Wolf [2, 3] and Rubin et al. [4]. *Stochastic modelling*—Markov chains, stochastic Petri nets, Monte Carlo simulation—has a long tradition in software reliability and process simulation [5, 6]. *Machine-learning-based prediction* dominates mining-software-repositories research on defects, effort, and outcomes [7, 8], and, in the business-process domain, predictive process monitoring [9]. Combining them is attractive because each covers a weakness of the others: PM provides an empirically grounded process structure, stochastic models turn that structure into probability distributions over future paths and durations, and ML can correct systematic errors using context features.

Related secondary studies. Existing secondary studies each cover one side of this space. In business process management, predictive process monitoring has been surveyed and benchmarked for outcome prediction [10], remaining-time prediction [11], and in general [9], but on business event logs rather than software development processes. General mappings of process mining applications [12, 13] treat software engineering as one domain among many and do not address forecasting. Within software engineering, secondary studies have covered process simulation modelling [14, 6], process mining of FLOSS event logs [15], software process mining with Petri nets [16], continuous software monitoring backed by process mining [17], repository mining [7], and defect prediction [8]. Four of these ([14, 15, 16, 17]) fall within our scope and were retained as included studies under EC4 (Table 6), which excludes only secondary studies that do *not* address PM or stochastic methods in software engineering. None of them maps the joint use of process mining, stochastic modelling, and forecasting on software development data. This study does.

Why the mapping matters. For the three communities named above, the mapping delivers a corpus of 456 records at their intersection, 380 of them with at least one family mark and 76 known only by their title, classified into four technique and data families on named cuts, with the one study that joins all three families identified. For the methodology of secondary studies, the review is also a deployment of LLM-assisted screening whose selection error we measure against a pre-registered human reference standard, following LLM4SCREENLIT [18]. The screen fell short of the pre-registered bar, and the paper names the mechanisms by which it did and quantifies what the human re-screen recovered. The mapping protocol itself was not pre-registered. The human check of the screen was pre-registered before the round-2 judgements.

Study design and contributions. We report a systematic mapping study against SEGRESS [1], following the mapping guidelines of Petersen et al. [19], the evidence-based software engineering guidelines of Kitchenham and Char-

ters [20], and the snowballing procedure of Wohlin [21]. The review addresses three research questions, each decomposed into two sub-questions. Table 1 names the output of each one: a family-code count on the cut named in the cell, or a field left unreported. Its contributions are (i) a pre-registered measurement of the selection error of an LLM-assisted screen (Recall 0.630 on the working-set sample and 0.725 on the auxiliary tier, against a bar of 0.90), with the failure mechanisms named and the human re-screen that the pre-registered rule triggered, which raised auxiliary Recall from 0.563 to 0.725 and added 150 studies; (ii) a corpus of 456 records at the PM, stochastic-modelling, and forecasting intersection (380 with at least one family mark; 76 known only by their title), classified into four technique and data families (TF-PM, TF-ST, TF-FC, TF-MSR) and into integration levels L0–L3, with the counts reported on named cuts (F1, F3, F4) and two free-text fields withdrawn after the extraction check (F2, F5); and (iii) the SPMF taxonomy and a four-direction research agenda proposed by the authors. Section 4.1 states once what the family-code counts are and are not. The mapping was motivated by, and is used to position, the authors' own integrated-pipeline proposal (PATHCAST). PATHCAST itself is not evaluated here; L3 is defined without reference to its components (Section 5.2), and PATHCAST appears as one research-agenda hypothesis among others.

Section 2 describes the protocol, including the eligibility criteria (Section 2.3); Section 3 the search and selection; Section 4 the results per research question (Sections 4.5–4.7), with the PDF-based spot-check of the extraction in Section 4.4; Section 5 the coded counts, the integration levels and the SPMF taxonomy (Section 5.2), and the agenda with its implications for practitioners (Section 5.3); Section 6 threats to validity, including the human validation of the selection; and Section 7 concludes.

2. Review Protocol

2.1. Research Questions

The questions below name the cuts on which family codes are reported. Two cuts in this paper are almost the same size (151 and 150 studies) and are disjoint, so they carry fixed labels throughout: **WS-151** is the 151 working-set studies that remain from the 169-study extraction (169 minus 16 removed studies and minus 2 extra identifiers of one paper), the cut on which the working-set family codes are reported; **ADD-150** is the 150 studies added to the corpus in this revision after the human re-screen (168 after de-duplication, minus 18 removed in the eligibility pass). No study belongs to both (Table 14).

2.2. Search Strategy

The search strategy combines automated database search with backward and forward snowballing. The automated search targets five digital libraries: IEEE Xplore, ACM Digital Library, Scopus, Web of Science, and SpringerLink. Search-string construction follows the PICO framework adapted for software engineering (Table 2): the population is the software engineering context, the intervention is the analytical technique, and the outcome is the forecast or process model; there is no comparison facet. From that decomposition three complementary term lists were derived,

Table 1: Research questions and the output this revision reports. A family code is the family mark on the cut named in the cell (Section 4.1). "Not reported" marks a free-text field withdrawn after the extraction check (Section 6) or a field not recomputed in this revision.

RQ	Sub-RQ	Question	Output in this revision
RQ1	RQ1.1	Which SDLC phases have been analyzed with process mining?	Not reported (field not recomputed).
RQ1	RQ1.2	What event log sources are used and how are they constructed?	Not reported (free-text fields, withdrawn).
RQ2	RQ2.1	Which process discovery algorithms have been applied to SDLC?	Algorithm names not reported (free text, withdrawn). TF-PM on WS-151: 104.
RQ2	RQ2.2	Which predictive and stochastic techniques have been used?	TF-ST on WS-151 (49) and on the 293 (152); technique tokens by family in Figure 5.
RQ3	RQ3.1	What is the level of integration between PM and stochastic methods?	L0–L3 on WS-151: 122, 28, 1, and 0 (Table 15).
RQ3	RQ3.2	What gaps exist for process-aware forecasting in the SDLC?	Research agenda RA1–RA4 (Section 5.3), proposed by the authors.

a principal string joining population and intervention, a complementary string for studies that use MSR terminology without canonical process-mining terms, and a frontier string for the intersection of stochastic methods with process mining in software contexts. The three are given next, followed by the database-specific strings that were run.

Principal String ($P \wedge I$). Block P captures the software engineering context:

```
("software engineering" OR "software development" OR "software process" OR "software
lifecycle" OR "software life cycle" OR "SDLC" OR "development lifecycle" OR "development
life cycle" OR "DevOps" OR "CI/CD" OR "continuous integration" OR "continuous delivery"
OR "agile development" OR "agile process" OR "issue tracking" OR "bug tracking" OR
"version control" OR "code review" OR "pull request" OR "software project" OR "software
repository" OR "software maintenance" OR "software evolution")
```

Block I captures the process mining and stochastic modeling techniques:

```
("process mining" OR "process discovery" OR "conformance checking" OR "workflow mining"
OR "predictive process monitoring" OR "process prediction" OR "remaining time
prediction" OR "process forecasting" OR "event log" OR "process simulation" OR "Monte
Carlo simulation" OR "Markov chain" OR "Markov model" OR "stochastic process model" OR
"transition probability" OR "absorbing chain" OR "lead time prediction" OR "cycle time
prediction")
```

Table 2: PICO decomposition for search string construction.

Facet	Role	Concepts
Population	Application domain	Software engineering, SDLC, software process, DevOps, CI/CD
Intervention	Analytical technique	Process mining, Markov chain, Monte Carlo, predictive PM, ML
Comparison	N/A	(omitted)
Outcome	Result	Forecasting, prediction, lead time, cycle time, process model

Complementary String (MSR → Process Modeling). Captures studies using MSR terminology without canonical process mining terms:

```
("mining software repositories" OR "software repository mining" OR "GitHub mining" OR
"Jira mining") AND ("process model" OR "Markov" OR "Monte Carlo" OR "stochastic" OR
"lead time" OR "cycle time" OR "throughput" OR "transition matrix")
```

Frontier String (Stochastic ∧ PM ∧ SE). Targets the intersection of stochastic methods with process mining in software contexts:

```
("Monte Carlo" OR "Markov chain" OR "stochastic simulation" OR "probabilistic forecast"
OR "transition matrix") AND ("process mining" OR "event log" OR "workflow mining" OR
"process model" OR "business process") AND ("software development" OR "software
engineering" OR "software process" OR "DevOps" OR "SDLC" OR "issue tracker" OR "commit
log" OR "build pipeline")
```

The blocks above are the shared term lists. They are not the Scopus syntax that was run. Table 3 is the ledger of every export in the corpus of 8,347 records: database, query name, and raw count. Every row was retrieved on 12 April 2026. The six Scopus rows sum to 2,377. All six Scopus strings, the four Web of Science strings, the two ACM query strings, and the four IEEE protocol strings, as executed, are reproduced in Appendix A. `scopus_principal` returned 1,595 records. None of the strings in the ledger contains a machine-learning block.

The eligibility window (IC1) runs from 1 January 1994 to the search execution date; publications dated 2026 are therefore partial-year coverage (indexed by 12 April 2026) and are reported as such. On 31 October 2026 a fourth string, with the terms *machine learning*, *deep learning*, *neural network*, *random forest*, *gradient boosting*, *xgboost*, or *lstm*, and with lead time, cycle time, remaining time, throughput, or just-in-time defect as the target, was run on Scopus. It returned 165 records. Title screening against the criteria in Table 6 left 19 includes, all under IC4c, and 2

Table 3: Search ledger. Each row is one export retrieved on 12 April 2026. The rows sum to 8,347.

Database	Query	String	Raw
Scopus	scopus_principal	Appendix A	1,595
Scopus	scopus_complementar	Appendix A	48
Scopus	scopus_markov_testing	Appendix A	498
Scopus	scopus_stochastic_petri	Appendix A	6
Scopus	scopus_simulation_bridge	Appendix A	200
Scopus	scopus_fronteira	Appendix A	30
Web of Science	wos_principal	Appendix A	632
Web of Science	wos_complementar	Appendix A	338
Web of Science	wos_msr	Appendix A	26
Web of Science	wos_fronteira	Appendix A	1
ACM DL	acm_principal	Appendix A	24
ACM DL	acm_msr	Appendix A	32
ACM DL	acm_extra_refs	supplementary BibTEX export	40
IEEE Xplore	ieee_manual_01–17	17 batches; strings in Appendix A	1,386
SpringerLink	springer_1	`"process mining" "software development"`	527
SpringerLink	springer_2	`"process mining" "software process"`	167
SpringerLink	springer_3	`"process mining" "software engineering"`	743
SpringerLink	springer_4	`"Markov chain" "software process"`	743
SpringerLink	springer_5	`"Monte Carlo" "software process"`	70
SpringerLink	springer_6	`"mining software repositories" "process model"`	140
SpringerLink	springer_7	`"stochastic" "software process" "event log"`	46
SpringerLink	springer_8	`"predictive process monitoring" "software"`	16
SpringerLink	springer_9	`"conformance checking" "software development"`	132
SpringerLink	springer_10	`"workflow mining" "software"`	300
Snowballing	snowball_both	one iteration; labels do not split direction	595
Control set	control_papers_bib	validation only	12
Total			**8,347**

records already in the 456. None of the 19 joins process mining, a stochastic model, and forecasting. They are not added to the 456: the screen used the title only, and the other four databases were not re-run. F5 is withdrawn, so those 19 are not used as a finding. Where a database supported a date restriction, the query used publication years 1994–2026; the Web of Science export was not date-restricted at query time, and IC1 is enforced at screening for every source (Table 4, note).

Table 4: Database-specific search adaptations and raw retrieval counts.

Database	**Search Field**	**Strings**	**Filters at query time**	**Records**
Scopus	TITLE-ABS-KEY	6	ar, cp; EN; PUBYEAR 1994–2026	2,377
IEEE Xplore	All Metadata	17 batches	Conf. + Journals; 1994–2026	1,386
Web of Science	TS (Topic)	4	Article, Proc.; EN; no date filter	997
SpringerLink	Simplified	10 queries	Chapter, Article; EN	2,884
ACM DL	Abstract	3	Date: 1994–2026	96
Snowballing	–	–	Backward + forward	595
Control set	–	–	Manual	12
Total				**8,347**

Note: all searches executed on 12 April 2026. After deduplication, 7 records dated before 1994 remained in the pool (6 from Web of Science, 1 from ACM DL). IC1 (1 January 1994 to the execution date) is enforced at screening; two pre-1994 auxiliary-tier records that had been confirmed in R1 were removed in this revision (Appendix B).

Snowballing procedure. Snowballing follows Wohlin [21] for one iteration. The identification file stores 595 records, each labeled as coming from both the reference list and the citation list of one source DOI. Those labels name 400 distinct source DOIs. The file does not split the 595 into a backward count and a forward count, and it does not record a second iteration. The stop rule is that single pass: a candidate is kept when the title or the abstract contains one process-mining or stochastic term and one software-engineering term. The file does not name an evaluator, does not give a count per source DOI, does not record how duplicates were removed inside the iteration, and was not restarted from the studies later marked as included. The ten papers in Table 5 check whether the database strings retrieve known studies. They are not the 400 source DOIs. The database searches are dated 12 April 2026. The snowball file does not carry a separate date.

Control-set check of the search strings. The search strings are checked against a control set of 10 papers that are known to be relevant to the review scope (Table 5). The ten papers were known when the strings were written. Retrieving all ten shows that the strings can return these known papers. It does not measure how many relevant studies the strings miss, and it is not recorded as a pass.

Table 5: Control papers for the search-string check.

#	Paper	Justification	Captured via
V1	Cook & Wolf (1998)	Seminal PM + SE	ACM
V2	Rubin et al. (2007)	PM framework for SW processes	Scopus
V3	Poncin et al. (2011)	PM + MSR intersection	IEEE
V4	Caldeira & Cardoso (2019)	PM + SW process assessment	IEEE
V5	Lemos et al. (2011)	PM for SW process management	IEEE
V6	Whittaker & Thomason (1994)	Markov + SW testing (seminal)	Scopus
V7	Tax et al. (2017)	PPM seminal (BPM baseline)	Control
V8	Rogge-Solti & Weske (2015)	Stochastic + PM forecasting	Scopus
V9	Rozinat et al. (2009)	PM → Simulation bridge	Control
V10	Gupta & Sureka (2017)	PM multi-repository in SE	ACM

2.3. Inclusion and Exclusion Criteria

Table 6 lists the inclusion and exclusion criteria. The unit to which they apply is fixed first, because it decides how duplicate and extended reports are counted.

Unit of counting. The unit of this mapping is the publication, as in Petersen et al. [19]; "study" in this article means one published report. Duplicate records of the same publication (same DOI, or same title under a different identifier) are merged during de-duplication. A conference paper and its later journal or conference extension are different publications and are both retained when each satisfies the criteria independently, because the extension may add evaluation the earlier report lacks and the mapping records what each report contains. Two such pairs exist in the corpus of 456: `b0f56a95` (ICSSP 2016) with `4fd2327e` (JSEP 2017), and `ae9983fa` (ISSREW 2022) with `d65695c6` (RAMS 2023). Counting studies rather than publications would give 454; both pairs carry the same family codes, so no proportion reported below moves by more than one percentage point (on WS-151, TF-ST would be 48 of 150 and TF-ST∩TF-FC 17 of 150; the 293-study joint cut would be 291).

The inclusion criteria are organized in two tiers. Formal criteria (IC1–IC3) filter on publication attributes and are applied during the database search configuration. Content criteria (IC4a–IC4d) require that the study's research contribution addresses at least one of four specific technique-domain intersections; they are applied during title/abstract screening and during the second-pass eligibility screen. The replication files label that second pass "FT". For almost every record the second pass read the title and the abstract. "Full text" in this article means a PDF was read. Exclusion criteria (EC1–EC5) are applied cumulatively across both screening phases. Criteria EC1–EC4 are evaluated by the

LLM-assisted screener; EC5 (full text inaccessible) is applied programmatically after the inter-library-loan recovery pass for Band D/E papers fails to retrieve a copy. In the auxiliary tier, EC5 is extended to cover non-recoverable metadata insufficiency: papers for which no usable abstract could be recovered after the multi-source enrichment cascade are flagged for human review in the revised protocol. In the **original execution run**, papers for which the primary and verifier raters produced an irreconcilable decision without accessible full text were closed under EC5 rather than escalated to human arbitration; Section 6 quantifies the resulting Lost Evidence against the pre-registered human check. That check is a two-rater sample. It is not a double screen of every record. The revised protocol treats T/A `maybe` and FT `pending` as positive referrals (LLM4SCREENLIT [18] R6).

At least one of IC4a–IC4d must be satisfied for inclusion. Criterion IC4b ensures that seminal works applying Markov chains to software processes—such as Whittaker and Thomason [5]—are captured despite predating the term "process mining." To avoid colliding with the publication filters IC1–IC3, Results and Appendix A report the four content families as **TF-PM**, **TF-ST**, **TF-FC**, and **TF-MSR** (IC4a–IC4d, respectively).

2.4. LLM-Assisted Screening Methodology

Given the scale of the working set (2,340 papers) and the requirement for systematic, reproducible screening decisions with traceable rationale, the title/abstract screening phase employs a language model as the primary screener. Specifically, *claude-haiku-4-5-20251001* is used via the Anthropic Message Batches API. The screening, QA, extraction, and verifier calls pass the model identifier and `max_tokens` and do not pass a temperature. The Messages API then uses its default temperature of 1. The prompt encodes the IC4a–IC4d and EC1–EC4 criteria (EC5 is applied programmatically post-screening), a short scope statement of the target intersection, and decision rules for the three-class output: `include`, `exclude`, or `maybe`. The scope statement was labelled with the name of the authors' research programme (PATHCAST); its content restates the four content families IC4a–IC4d and adds no further criterion, but we report it because a framework label in the prompt is a potential source of selection bias (Section 6). The exact prompts are in the replication package.

The screening prompt instructs the model to (i) identify which ICs are satisfied, (ii) identify which ECs apply, (iii) assign a decision with a one-to-two sentence rationale, and (iv) return the result as structured JSON. The `maybe` class captures genuine uncertainty—typically arising from missing abstracts or ambiguous domain terminology—without forcing a binary decision.

At the second-pass eligibility screen, the same model is applied with a stricter prompt that enforces a binary outcome (`include` or `exclude`) with `pending` reserved only when the abstract is genuinely insufficient for a decision. The FT prompt also instructed the model to prefer `exclude` when undecided between `include` and `exclude`; this rule biases the FT stage toward false negatives, which is the error the human validation in Section 6 is designed to measure.[4]

[4]The two protocols use different nomenclature for the uncertainty class: T/A uses `maybe` (epistemically broad: criterion ambiguity *or* abstract

Table 6: Inclusion and exclusion criteria.

Type	ID	Criterion
Inclusion	IC1	Published between 1 January 1994 (from Whittaker & Thomason onward) and the search execution date (12 April 2026).
	IC2	Language: English.
	IC3	Peer-reviewed journal article, conference paper, or workshop paper (≥4 pages).
	IC4a	Applies process mining techniques (discovery, conformance checking, predictive process monitoring) to software development artifacts (commits, issues, pull requests, CI/CD logs, bug trackers, VCS).
	IC4b	Uses stochastic models (Markov chains, Monte Carlo, stochastic Petri nets, absorbing chains) to analyze software development processes or workflows.
	IC4c	Proposes or evaluates forecasting of software process metrics (lead time, cycle time, remaining time, throughput, defect rates) from event logs or repository data.
	IC4d	Mines software repositories (GitHub, Jira, VCS, CI/CD platforms) to discover, analyze, or improve software process models.
Exclusion	EC1	Application domain is exclusively outside software development (healthcare, manufacturing, supply chain) without demonstrated relevance to SDLC processes.
	EC2	Software appears only as the implementation platform; the process being analyzed is not a software development process.
	EC3	Purely algorithmic or theoretical contribution without empirical evaluation in a software development context.
	EC4	Secondary study (SLR, mapping study, survey) not specifically addressing PM or stochastic methods in software engineering.
	EC5	Record cannot be judged on the evidence obtainable. Three sub-causes are distinguished for PRISMA reporting: **EC5a**, full text sought individually (open access, institutional, ILL, author-provided preprint) and not retrieved; **EC5b**, no abstract recoverable after the multi-source enrichment cascade and no PDF from the automated recovery pass, so the decision would rest on the title alone; **EC5c**, abstract or PDF available but the decision-critical information is absent or the two model raters stay irreconcilable. Applied programmatically after Band D/E recovery and auxiliary enrichment.

This approach extends prior work on LLM-assisted systematic reviews [22, 23, 18] to the software process mining domain. The primary screener produces structured outputs that enable post-hoc quality analysis. Validation follows a three-stage protocol: (i) a pre-registered two-rater check of a stratified random 20% sample, blind to the LLM (n = 468 at title/abstract, n = 177 at the second pass), which is the source of the Recall figures (Tables 17 and 18); (ii) cross-model verification on a separate stratified 20% sample, with *claude-sonnet-4-6* as second rater and Cohen's κ reported following Landis and Koch (1977) [24]; and (iii) a two-reviewer re-screen of the working-set records the LLM had excluded and of the auxiliary records that were pending or otherwise unclassified. On that re-screen the two reviewers agreed with 92% of the suggested labels, including the labels that a heuristic pass had proposed for the later S1 batches and for all of S3. The sheets for that re-screen (4,005 rows) store a decision equal to the suggestion on every row, so the labels they did not accept are not marked there. Rebuilt from the title or abstract the LLM treated as sufficient, that share of the coded includes is 32 of 338 (9.5%), listed in `labels_ta_insufficient_for_include.csv` and removed in Section 4. Cross-model verification is a secondary signal. The Recall figures come from the sample, not from the 92% agreement.

Operationally, the **original execution run** did not preserve an open-ended manual-arbitration queue: unresolved `maybe`/`pending` cases were closed under EC5-extended. The **revised protocol** routes such cases to human review instead. In the working set, T/A `maybe` decisions were forwarded to FT screening, so no human-relevant record in the T/A validation sample was lost at T/A (Section 6).

2.5. Reporting-Quality Assessment

Under SEGRESS [1] §4.3.2, risk-of-bias assessment is optional for mapping studies; we retained a **reporting-quality** checklist (not a risk-of-bias instrument) adapted from Dybå and Dingsøyr [25], with eight binary criteria. It is *not* a selection filter, and the 4/8 line is not applied to any finding. The same model that screens and extracts (*claude-haiku-4-5-20251001*) also assigns the eight scores. A score produced from an abstract is an abstract-only score. It is reported as such and is not pooled with a score produced from a PDF. The checklist is descriptive. It is not a validity judgement on primary-study outcomes (Table 7):

The eight scores are descriptive. The same model that screens and extracts assigns them. No finding uses a cutoff on the sum.

QA scoring is performed by the same LLM-assisted protocol used for screening (Section 2.4): *claude-haiku-4-5-20251001* receives a strict rubric for each criterion and returns a binary score with a brief justification. The full prompt and the per-study scores are part of the replication package (Section 6.4). Table 8 reports the scores for the WS-151 cut (the 151 working-set studies that remain; Section 2.1).

insufficiency), whereas FT uses `pending` (narrowly: abstract insufficiency only). The semantic difference is intentional — the T/A pass operates on metadata where ambiguity is expected, while the FT pass commits to binary decisions whenever the abstract permits and escalates only the abstract-insufficient subset to manual full-text retrieval. This mapping is preserved in the cross-model κ analysis (Tables 19 and 20).

Table 7: Quality assessment checklist.

#	Criterion	Score
QA1	Are the research objectives clearly stated?	0/1
QA2	Is the software engineering context described in detail?	0/1
QA3	Is the data source (event log) described reproducibly?	0/1
QA4	Is the PM or stochastic technique formally defined?	0/1
QA5	Are results validated empirically?	0/1
QA6	Are threats to validity discussed?	0/1
QA7	Are data and/or code publicly released?	0/1
QA8	Are process model quality metrics reported (fitness, precision)?	0/1

Table 8: Quality scores for the WS-151 studies (the 151 working-set studies that remain). The scores are descriptive and are not a filter.

Score band	Count	Notes
Studies assessed	151	WS-151: working-set studies still in the corpus. Not the corpus of 456, and not the 150 added studies (ADD-150).
Studies with score $\geq 4/8$	124	Score assigned by the model. Not a filter.
Studies with score $< 4/8$	27	Not a filter.
Mean QA score	4.90	Mean and standard deviation (1.36)
Median QA score	5.00	Median and interquartile range (4.00–6.00)

The model scored 27 of the WS-151 studies below 4/8. The score is not a filter. Section 5.1 does not apply a cut on 4/8.

2.6. Data Extraction

For each included study, the attributes in Table 9 are extracted by the same model that screens and scores the studies (Section 2.4), from the title and abstract for every study and from the PDF where one was read. Three groups of fields serve different purposes. The IC combination (IC4a–IC4d) yields the four technique and data families and, from them, the integration level L0–L3; these are the fields on which every count in Sections 4 and 5 rests. The categorical fields (study type, stochastic method, validation type) are reported descriptively. The free-text fields (event log source and construction, specific algorithms, tool, dataset size, model quality) are the ones checked against a human rater on 31 studies and withdrawn as F2 and F5 (Section 6); no percentage in this article is computed from them.

Table 9: Data extraction form.

Field	**Description**
ID	Unique identifier
Authors, Year, Venue	Bibliographic metadata
SDLC Phase(s)	Lifecycle phases covered
Event Log Source	Jira, GitHub, GitLab, Jenkins, etc.
Event Log Construction	Manual, semi-automatic, or automatic
PM Technique Category	Discovery / Conformance / Enhancement / Prediction
Specific Algorithm(s)	Alpha, Heuristics, Inductive Miner, etc.
Stochastic Method	Markov / Monte Carlo / Stochastic Petri Net / None
ML Technique	RF, XGBoost, LSTM, etc. / None
Prediction Target	Remaining time, outcome, lead time, delay, etc.
Integration Level	L0–L3 (derived from technique families; Table 15)
Validation Type	Case study / Experiment / Industrial / Simulation
Tool/Platform	PM4Py, ProM, Disco, custom, etc.
Dataset Size	Number of events and cases analyzed
Process Model Quality	Fitness, precision, generalization (if reported)

3. Search Results and Study Selection

3.1. Retrieval, Deduplication, and Working Set Construction

The database rows in Table 3, apart from snowballing and the control set, returned 7,740 records. One snowballing iteration, stored as 595 records attributed to 400 source DOIs, yielded the additional records (Section 2.2).

Together with 12 control papers maintained as a validation-only set, the initial corpus totaled 8,347 records. Table 10 summarizes the retrieval by source.

Table 10: Record retrieval by source. The second column counts each of the 5,783 unique records once, under the source that supplied the retained copy after cross-source deduplication.

Source	**Raw Records**	**Unique after cross-source dedup**
Scopus (6 queries)	2,377	1,852
SpringerLink (10 queries)	2,884	1,543
IEEE Xplore (manual)	1,386	1,263
Snowballing	595	575
Web of Science (4 queries)	997	514
ACM Digital Library (2 queries + 1 export)	96	34
Control set (validation only)	12	2
Total	**8,347**	**5,783**

Cross-source deduplication removed 2,564 duplicate records (30.7%), yielding 5,781 unique primary-corpus papers plus 2 unique control papers. Deduplication used normalized DOI matching, followed by fuzzy title matching (Levenshtein similarity ≥ 0.9) for records without DOIs.

Screening the 5,783 records in a single pass would require substantial resources. A working set of 2,340 papers was therefore constructed from the highest-signal primary queries—those whose query design directly targets the review intersection (process mining, stochastic modeling, and software development)—as shown in Table 11. Of the 2,340 working-set identifiers, 1,976 are among the 5,783 deduplicated records, the other 364 rows are the same papers under a different identifier inside the 3,807 (5,783 – 1,976), the union of the two identifier lists is the 5,783, and 345 of those 364 are Scopus rows in the working-set file, so 2,197 − 345 = 1,852, the Scopus count in the deduplicated file. Of the 364 pairs, 69 appear in the corpus of 456 under the working-set identifier and 30 under the canonical identifier. No pair appears under both identifiers. The 3,807 is the auxiliary identifier list that was screened.

Of the ten known control papers, eight are retrieved by the primary queries and all ten appear once a validation-only set is counted. That count does not measure search recall.

3.2. Title/Abstract Screening

All 2,340 working-set papers were screened against IC4a–IC4d and EC1–EC4 using the LLM-assisted protocol described in Section 2.4. The screening was executed across five sequential batches of up to 500 papers each via the Anthropic Batches API, producing structured JSON outputs with decision, rationale, and matched criteria for each record.

Table 11: Working set composition.

Database	Queries included	Rows
Scopus	scopus_principal, scopus_complementar, scopus_fronteira, scopus_markov_testing, scopus_stochastic_petri, scopus_simulation_bridge	2,197
IEEE Xplore	ieee_manual_01 (PM + SE subgroup)	97
ACM DL	acm_principal, acm_msr	46
Total		**2,340**

Table 12: Title/abstract screening results.

Decision	Count	%
Include	111	4.7
Maybe	775	33.1
Exclude	1,454	62.2
Total	**2,340**	100.0

Table 12 gives the decisions. The `include` set (111 papers) and the `maybe` set (775 papers) together form the second-pass queue of 886 papers. The `exclude` decisions were driven primarily by EC1 (domain outside software development, principally generic business process management) and EC3 (theoretical algorithm proposals without SE evaluation).

Information basis at T/A. For 1,695 of the 2,340 working-set records (72.4%; mostly Scopus and ACM) the T/A decision was made on the title alone: the Scopus Search API leaves the abstract field empty for most records, and the separate Abstract Retrieval call was not part of the original extractor. This is one reason for the large `maybe` class. After the R1 review we recovered abstracts for 1,131 of these records and re-screened them with the same model, prompt, and decision rule; after discarding 44 recoveries that had attached the wrong abstract, 1,087 re-screenings remain, which raised working-set abstract coverage from 645 of 2,340 (27.6%) to 645 + 1,087 = 1,732 (74.0%). Of the 1,087, 372 (34.2%) changed decision and 4 moved from `exclude` to `include`/`maybe`. Those four records were not added as a separate batch. Studies that entered the corpus because a human later judged them eligible are the 168 described in Section 4, of which 18 are removed in the eligibility pass. The 150 that remain are ADD-150, the studies added in this revision; ADD-150 is disjoint from WS-151, the working-set cut on which the family codes are reported.

3.3. Second-Pass Eligibility Screening ("FT")

The 886 papers forwarded from the T/A phase entered the second-pass eligibility screen, using the strict prompt described in Section 2.4 (decisions limited to `include`, `exclude`, or `pending`; `maybe` is not permitted at this stage). To maximise the number of papers assessable by the LLM, a multi-source abstract-enrichment cascade was executed prior to screening—combining DOI-based lookups via Semantic Scholar, OpenAlex, CORE, and Crossref, followed by title-based fallback lookups—raising abstract coverage from 16.6% to 72.6% (643 of 886 papers).[5]

Information basis at the second pass. Despite its label, this stage was not a full-text reading for almost all records: the screening prompt received the same (enriched) title and abstract as T/A, under the stricter decision rule described in Section 2.4, and a PDF was read for only 0.5% of the 886 records. We keep the label "FT" for traceability with the replication package, but its decisions are abstract-based eligibility decisions. Full texts were read by humans only in the validation samples (Section 6). After R1, the same model screened 8 records with conflicting abstract-based decisions on the extracted PDF text (`ft_screened_by = llm_pdf` in the replication package). Three of those eight were `include` on the PDF. `ba2ff831` is in the 169-study tier and in the 456. `9188583f` (Setamanit et al., 2006) is in the 456 under its canonical identifier, `b5b1402b`, the snowballing record with the same DOI (10.1145/1138506.1138510); the pair is one of the 364 twin identifiers (Section 3), and only the canonical identifier is listed in Supplementary Material S1. `ee562777` (Bala et al., 2015) is in the 169 and in the 340. A first draft of this revision had placed it among the coded includes removed on the title and abstract (Section 4), with the note that the abstract does not fix a software development process. That note was written from the abstract. The PDF-based decision was `include` under IC4a and IC4d, and the paper's evaluation mines SVN and Git logs of open-source software projects. The rule that defines that removal list, an include whose title or abstract the model treated as sufficient, does not cover a PDF-based include, so the record was reinstated: the removal list has 32 studies, not 33, and `ee562777` is in the 456, in WS-151, and in the 293-study joint cut.

Table 13 summarises the outcome.

Among the 717 excluded papers, EC3 (algorithmic or theoretical contribution without SE evaluation, $n = 232$) and EC1 (domain outside software development, $n = 205$) jointly account for the largest share of FT-stage exclusions. EC5 (full text inaccessible despite all retrieval attempts, active post-audit $n = 145$) corresponds to the residual portion of the original 148-paper Band D/E stratum for which no open-access copy, institutional access, or author-provided preprint could be located; these are recorded in a dedicated traceability file and documented in the PRISMA flow as a named exclusion stratum. EC2 (software as implementation platform rather than analyzed process, $n = 110$) and EC4 (out-of-scope secondary studies, $n = 60$) account for the remainder.

Figure 1 presents the complete PRISMA 2020 flow diagram [26] of the selection process.

[5] Papers were processed in order of a five-band priority scheme: Band A (91 papers, T/A `include` ≥2 ICs), Band B (20, T/A `include` 1 IC), Band C (39, T/A `maybe` + abstract), Band D (152, T/A `maybe` + IC signal, no abstract), Band E (584, T/A `maybe`, no IC signal, no abstract). Abstract enrichment and LLM re-screening were applied to Bands A, B, C, and the subset of D/E that gained an abstract after enrichment.

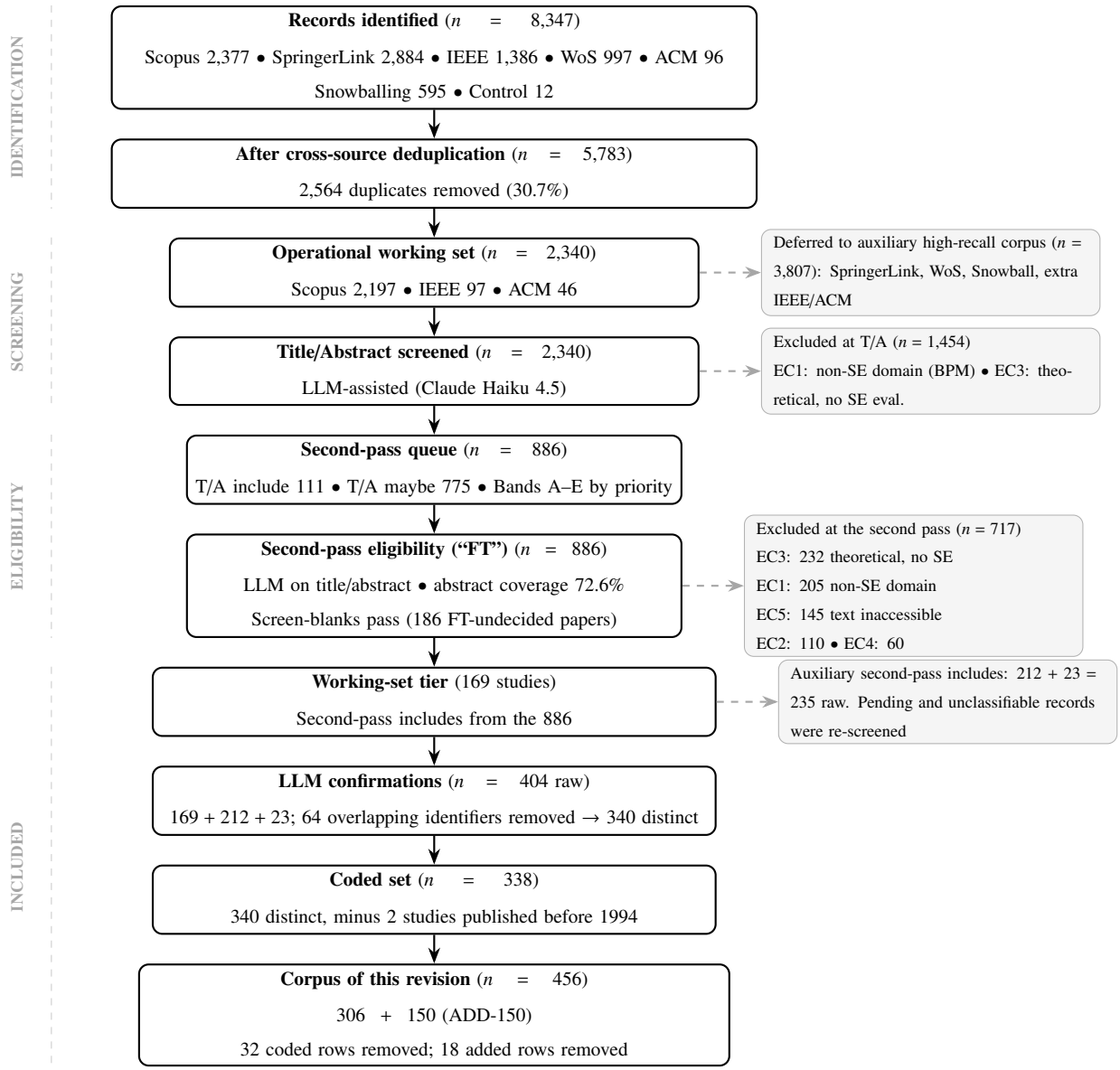


Figure 1: PRISMA 2020 flow of the study selection. The 2,340 and the 3,807 are identifier lists. They share 364 papers, and their union is the 5,783. The included phase continues through the auxiliary screen, the removal of overlapping identifiers, the removal of two pre-1994 studies, and the human re-screen. The mapped set is 456 studies (Table 14). The 318-study joint tables are an older cut (381 records minus 63 duplicates) and are not a node of this flow. In the original run, 595 records with no abstract and 16 records that stayed pending were closed under EC5; this revision re-screened those records.

Table 13: Second-pass eligibility outcomes (886 papers forwarded from the T/A phase). The replication files label this stage FT. The prompt received the title and the abstract.

Outcome	**Count**	**%**
Included	169	19.1
Excluded	717	80.9
EC3 (algorithmic, no SE eval.)	(232)	(26.2)
EC1 (non-SE domain)	(205)	(23.1)
EC5 (full text inaccessible, post-audit active stratum)	(145)	(16.4)
EC2 (SW as implementation)	(110)	(12.4)
EC4 (out-of-scope secondary)	(60)	(6.8)
Total	**886**	100.0

Note: EC counts are non-exclusive — a single paper may trigger multiple ECs when more than one applies (e.g., a non-SE-domain paper whose contribution is purely theoretical). The EC5 count of 145 is the active post-audit stratum after re-screening the original historical EC5 band of 148 papers and reclassifying 3 of them to EC3. By sub-cause (Table 6; rebuilt from `ft_evidence_status` and `ft_screened_by` in the replication package), the 145 are EC5a 6 (PDF sought manually and not found), EC5b 136 (no abstract and no PDF), and EC5c 3 (two abstracts too short to decide, one PDF read without a decision). The sum of per-EC counts is 232 + 205 + 145 + 110 + 60 = 752, which exceeds the 717 excluded papers by 35 (corresponding to papers triggering more than one EC). The percentages above are with respect to the 886-paper FT queue.

4. Results and Synthesis

4.1. Scope of the Counts

Every count in this section and in Section 5 is a count of family codes on a named cut of this corpus: WS-151, the 293 joint-table studies, or the 380 studies with at least one family mark. The codes on the 338 are those the model assigned in the original extraction, spot-checked on the 73-PDF subset; every coded study carrying two technique families (n = 99) was re-examined by the authors for the third family (PDF for 78, abstract for 21; perfect agreement on the 30 double-read rows, κ undefined because both raters used a single category); none moved from L1 to L2 and none reaches L3; the marks on the 74 added studies read from an abstract or a PDF were assigned by the authors. The family counts are lower bounds: under-marking on the L0 rows and on the 76 blank rows was not audited, and correcting it would raise, not lower, every family count; the 46 added studies first labelled "none of the four" and then assigned a family show that such under-marking occurs. Three consequences follow and are stated here once rather than at each count. First, the counts describe this corpus, not the literature: the screen missed eligible studies (Recall 0.630 and 0.725, Section 6), so a proportion on a cut is not a rate at which published work combines the techniques, and a rare pattern in a cut is not a rare pattern in the field. Second, two free-text families of fields (event-log construction and dataset availability, F2; process-derived features, F5) are withdrawn, because the exact match between the model and a human rater on 31 studies is 0%; no percentage from them is reported. Third, the SPMF taxonomy and the agenda RA1–RA4 are proposals by the authors, grounded in their reading of the cited studies, not

quantities derived from F1–F5. The 76 added studies read from the title alone carry no family mark and are outside every count below.

4.2. Overview of Included Studies

This revision includes **456 studies** (publications; Section 2.3): 306 that remain from the coded list of 338, plus the 150 added studies that remain after the eligibility pass (ADD-150). ADD-150 is not the WS-151 cut: WS-151 is the 151 working-set studies inside the 306 on which the family codes below are reported (Section 2.1), and the two sets share no study.

The 338 are the LLM list of 340 studies, minus two papers from 1991 and 1992. The human re-screen marked 188 records as eligible. Twenty of them repeated a study already in the list, leaving 168. Eighteen of those 168 fail an inclusion or exclusion rule and are removed. Seventeen had been read from an abstract or a PDF: the extraction either names no technique in IC4a–IC4d, or the text is outside software development (EC1 or EC2). One, identifier `46815264`, is dated 1992 and fails IC1. Forty-six others had been labelled "none of the four" even though the extraction names a technique; they stay, with that family. The 18 removals and the 46 assignments are listed in `corpus_eligibility_pass.csv`. Of ADD-150, 11 were read as a PDF, 63 from the abstract, and 76 from the title only.

Thirty-two studies, 32 of the 338 coded includes (9.5%), are then removed. The re-screen sheets do not mark them: on all 4,005 rows the decision equals the suggestion. The list was rebuilt from the title or abstract the LLM treated as sufficient. Each was an include for which that text does not support inclusion. The object is generic process mining or forecasting, a business process, product execution, or system architecture. Two of them, `96767878` and `f449ac41`, have no abstract, and the inclusion note itself asks for the full text. Sixteen come from the working set, seven from the first auxiliary tier, and nine from the second. The list is `labels_ta_insufficient_for_include.csv`. The coded studies that remain are 306, and the corpus is 306 + 150 = 456, where the 150 is ADD-150.

The three technique families were marked on **380** of the 456: the 306, plus the 74 added studies that had an abstract or a PDF and meet at least one content criterion. One of those 380, PRIMAD, has all three families. The 76 studies known only by the title were left blank. A blank is not counted as a no.

Table 14 is the single count of every cut used in the paper. Four of those rows answer different questions, and they are not interchangeable. The context set is the 456. The 32 removed from the coded list are the LLM includes whose title or abstract does not support inclusion. The 74 added studies stay because the extraction names a technique in IC4a–IC4d. The 76 title-only studies were not marked, and a blank is not treated as satisfying a content criterion. The rows with technique-family codes filled before this round were the 338; 306 remain after the removal of 32. The rows with at least one family mark, including the 74 added studies, are the 380. The intersection of process mining, a stochastic model, and forecasting is one study inside that 380. Each research question is answered on the cut named in its sentence. The working-set family codes use WS-151. That is the 169 extraction minus 16 studies whose title or abstract does not support inclusion, and minus two extra identifiers of one paper the corpus keeps once (`2647ea07`).

WS-151 lies inside the 306 and contains none of ADD-150. The joint family codes use 293 studies. That is the 318 minus those 16, minus seven studies from the first auxiliary tier removed for the same reason, and minus the two pre-1994 papers. Nine of the 32 removed studies are not in the 318. The joint tables use 318. That 318 is what remained after 63 duplicate records were removed from an earlier combined list of 381. The 340 is what remained after 64 duplicate records were removed from 404 raw confirmations. The two cuts were built by different removals. Adding the 23 studies of the second auxiliary pass to 318 does not produce 340. The 32 removed above are the coded includes whose title or abstract does not support inclusion. Appendix B groups the 456 studies by family.

The list is `included_studies_506.csv` (456 rows; the filename keeps an earlier count). The file of 340 rows is kept so the removal of the two pre-1994 papers can be checked. Thirty-five of the ADD-150 studies were assigned a model score of at least 4/8. A score taken from an abstract is not a quality assessment of the study, and no score is used to keep or drop a study. F2 and F5 are withdrawn, and their percentages are not reported. The percentages under F1, F3, and F4, apart from the count of one study in 380, are the family codes on WS-151 and the 293. On the original 169, TF-PM with neither TF-ST nor TF-FC is 105, not 68: the 68 are the code that is exactly TF-PM, and 37 more are TF-PM with TF-MSR only. On the original 318 the same definition is 138, not 78: the 78 are TF-ST together with TF-FC and no TF-PM. The figures below use WS-151 and the 293. Their status is stated once in Section 5.1.

The WS-151 studies span 1995–2026 (publications dated 2026 that were already indexed on 12 April 2026 are retained), with a clear acceleration from 2014 onward (Figure 2). One hundred and five of them (69.5%) appeared in 2015–2026. A completed SEGRESS [1] reporting checklist is in Appendix C.

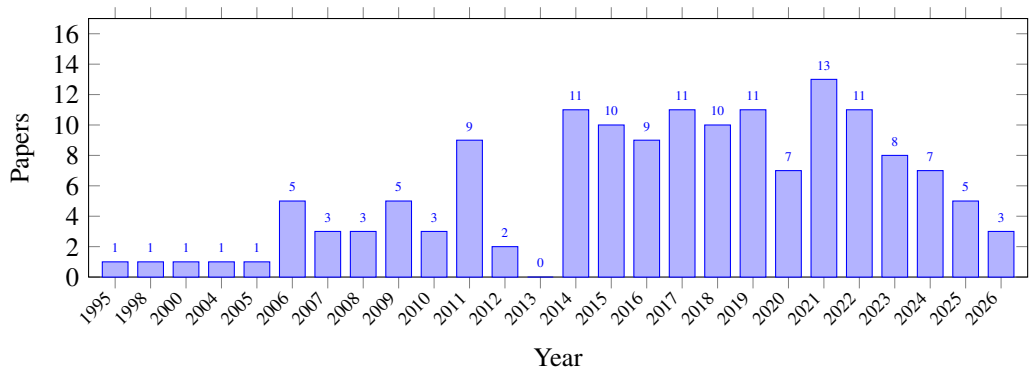


Figure 2: Publication year of the WS-151 studies (the 151 working-set studies that remain).

By document type, the majority are conference papers and journal articles. On the WS-151 cut, the source database is Scopus for 105 rows, IEEE Xplore for 33, and ACM DL for 13. On the cut of 293 the second source is Web of Science (74), ahead of IEEE Xplore (57). The extraction file is the earlier 169-row working-set extraction, produced by a structured LLM-assisted pipeline (*claude-haiku-4-5-20251001*) on locally available PDFs (75/169, of which 73 were operationally coded in the triangulation subset) or abstract/title metadata (94/169). Nineteen of those 169 rows are not in WS-151.

Table 14: Counts used in this paper. Each row is a different cut.

Studies	**Meaning**
8,347	Records retrieved, before deduplication.
5,783	Records after deduplication.
2,340	Working-set identifier list. 1,976 of these identifiers are in the 5,783.
3,807	The other identifiers in the 5,783 (5,783 − 1,976). This list holds the second identifier of 364 papers that are also in the working set. The union is 5,783.
404	Raw LLM confirmations (169 + 212 + 23).
340	Distinct LLM list (404 minus 64 duplicate records).
338	The 340 minus two papers from 1991 and 1992.
381	Earlier combined list, before the 318 cut.
318	Joint tables (381 minus 63 duplicate records). Not 340 minus 23.
WS-151	Working-set family codes: 151 studies. The 169 minus 16 removed studies and minus 2 extra identifiers of one paper. A subset of the 306.
293	Joint family codes. The 318 minus 23 removed studies and minus 2 pre-1994 papers.
32	Coded includes removed (32 of 338, 9.5%). The title or abstract does not support inclusion. Not marked on the 4,005-row sheets.
306	Coded studies that remain (338 minus 32).
ADD-150	Studies added in this revision: 150. Of 168 after de-duplication, 18 were removed in the eligibility pass. Disjoint from WS-151.
456	This revision (306 + ADD-150).
380	At least one family mark (306 + 74); the denominator of the triple count.
76	Added studies known only by the title. Marks left blank.

4.3. Criterion Coverage and Exclusion Patterns

Figure 3 shows the distribution of inclusion criteria among the WS-151 studies (criteria are non-exclusive; a paper may satisfy multiple); the exclusion side of the same criteria is read from the 717 papers excluded at the second pass.

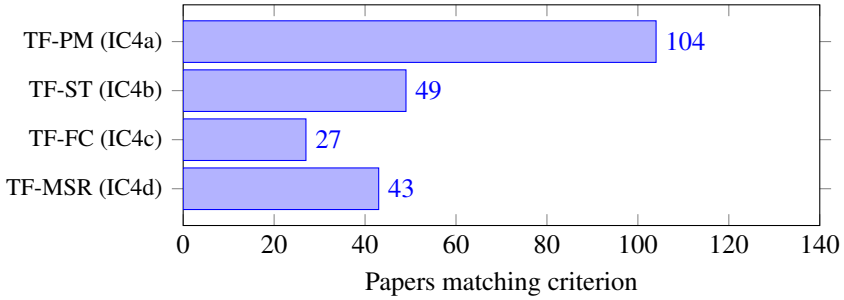


Figure 3: Inclusion criteria matched by the WS-151 studies (the 151 working-set studies that remain). Counts are non-exclusive. TF-PM, TF-ST, TF-FC, and TF-MSR are the content criteria IC4a–IC4d. TF-PM with neither TF-ST nor TF-FC is 92 of these 151.

On the WS-151 cut, 104 studies (68.9%) carry TF-PM, 43 (28.5%) TF-MSR, 49 (32.5%) TF-ST, and 27 (17.9%) TF-FC. TF-PM with neither TF-ST nor TF-FC is 92 rows (60.9%). The other 102 of WS-151 do not carry TF-ST, and 49 + 102 = 151. That split is the family-code count labeled F1 in Section 5.1.

Among the 717 FT-excluded papers (Table 13), the largest exclusion reason is EC3 (algorithmic or theoretical contribution without software-engineering evaluation, $n = 232$), indicating a substantial body of process mining and stochastic modeling research that has not been empirically applied to software engineering contexts. EC1 (domain exclusively outside software development, $n = 205$) is the second-largest reason and consists primarily of generic business process management (BPM) studies that use the same terminology as the search strings but apply techniques to organizational processes unrelated to SDLC. EC5 (full text inaccessible, active post-audit $n = 145$; original historical stratum $n = 148$) closes the group of large strata, followed by EC2 (software as implementation platform rather than analyzed process, $n = 110$) and EC4 (out-of-scope secondary study, $n = 60$). EC counts are non-exclusive: the sum of per-EC counts (232 + 205 + 145 + 110 + 60 = 752) exceeds the 717 excluded papers by 35, corresponding to papers that triggered more than one EC simultaneously. EC3 and EC1 are the two largest reasons in this exclusion list. That is a property of the list. It does not confirm the family codes in Section 5.1.

4.4. Full-Text Extraction and PDF-Based Triangulation

Structured data extraction was performed on the full 169-study set using the extraction template defined in Table 9. For each study, the LLM extracted the research question, study type, research contribution, PM technique, stochastic technique, software artifact, software process, data source, dataset availability, tool used, main finding, limitations, and replication-package status. Title-and-abstract metadata was used as the primary input for studies without locally available PDFs (94/169); for the remaining 75/169 studies a locally accessible PDF was available and used to triangulate against the LLM-extracted fields. Of these 75 PDFs, 73 were fully coded in the operational triangulation

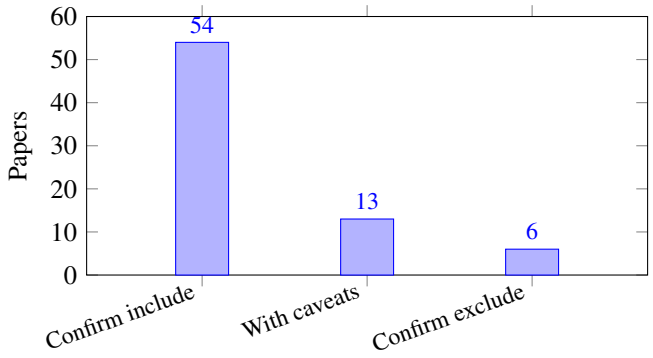


Figure 4: Reading of the 73 PDFs: 54 confirm-include, 13 confirm-include with caveats, and 6 confirm-exclude. These labels are not the screen decisions Include, Maybe, and Exclude. The PRISMA flow is unchanged.

spreadsheet used for the figures and positioning tables below.

The 73-paper PDF triangulation subset is therefore not a separate selection stage and does not override the PRISMA counts; it is a spot-check of the extracted fields, not a re-extraction. The human check of those fields on 31 studies (0% exact match on free text) and its consequences for F1–F5 are stated in Section 4.1.

Figure 4 summarizes the operational triage profile of the 73 coded PDFs (54 confirm-include, 13 confirm-include with caveats, 6 confirm-exclude after full-text inspection); the formal selection decisions reported in the PRISMA flow (Figure 1) are unchanged.

4.5. RQ1: Application Landscape

RQ1 asks where in the lifecycle process mining has been applied (RQ1.1) and on which event logs (RQ1.2). Both are answered here from the cited studies, because the two fields that would quantify them are not recomputed or are withdrawn (Table 1).

RQ1.1: SDLC phases covered. The phase field on the 169 extraction was not recomputed after 16 studies and two extra identifiers left that cut, so phase percentages are not reported in this revision; the family codes below use WS-151.

RQ1.2: Event log sources and construction. The source families, the public-dataset count, and the replication-package count are the free-text fields withdrawn as F2 (Section 4.1), so no percentage is reported for them. The studies cited below illustrate the construction routes the corpus contains.

A complementary body of work constructs Markov models directly from GitHub repository traces, bypassing explicit process discovery in favour of manual state-space definition. Jo et al. [27] modelled developer activities in GitHub repositories as Discrete-Time Markov Chains (DTMCs), finding that commits account for 62.73% of long-term activity, and employed probabilistic temporal logic to extract collaboration patterns relevant to project management. This approach was extended in [28], where Probabilistic Model Checking with Computation Tree Logic was

applied across five repositories to characterise temporal dynamics, state-transition probabilities, and key behavioural drivers, yielding transparent and explainable project forecasts. Ortu et al. [29] complemented these structural analyses by combining Markov chains with social network analysis to identify fault-insertion and fault-fixing behavioural patterns across Apache Software Foundation projects, demonstrating that developer expertise significantly moderates defect introduction likelihood. Collectively, these studies confirm that SDLC event logs extracted from public repositories are sufficiently rich to support state-transition modelling. However, the event-to-state mapping adopted across these works is defined manually and differs from study to study; none derives the state space from a discovered process model.

4.6. RQ2: Techniques and Prediction

RQ2 asks which discovery algorithms (RQ2.1) and which predictive or stochastic techniques (RQ2.2) the studies apply. Algorithm names are a withdrawn free-text field, so RQ2.1 is answered from the cited studies; RQ2.2 rests on the TF-ST family code (49 on WS-151, 152 on the 293) and on the technique tokens of Figure 5.

RQ2.1: Discovery algorithms applied to the SDLC. The canonical origin of process mining applied to software development contexts can be traced directly to the foundational contributions of Cook and Wolf in the mid-1990s. In their seminal work, [2] introduced three automated process discovery methods—algorithmic grammar inference, Markov models, and neural networks—demonstrating that finite-state machine representations of software processes could be extracted automatically from event traces, substantially lowering the barrier to formal process modeling. This line of inquiry was consolidated in [3], which refined the event-based discovery methodology and established the conceptual vocabulary of inferring workflow structures from historical execution data, a paradigm that remains central to the field today. Notably, both contributions embedded stochastic reasoning—particularly Markov chain representations—within the discovery pipeline, anticipating the probabilistic extensions that would later characterise mature process mining frameworks. The translation of these principles into an explicit, named framework for software engineering contexts was accomplished by [4], who applied the ProM toolkit to source code management logs and demonstrated multi-perspective analysis spanning control-flow, informational, and organisational dimensions. Together, these three works constitute the seminal arc of TF-PM, establishing automated discovery from SDLC event logs as both technically feasible and analytically productive.

The family-code count on WS-151, which this revision did not re-mark, has 104 studies (68.9%) matching TF-PM. The same count has 27 papers (17.9%) matching TF-FC. The names of individual algorithms are free-text fields. They were not re-extracted. Those names are not reported.

TF-ST is 49 of the WS-151 studies (32.5%). The other 102 do not carry that code, and 49 + 102 = 151. The column `stochastic_technique` is not that code. On these 151 rows it is `none` for 97 and a named technique for 54, and 97 + 54 = 151. Twenty-six of the 54 also contain the token `other`. Six rows coded TF-ST have `none` in the column, and 11 rows with a named technique are not coded TF-ST. On the original 169 the same column was `none`

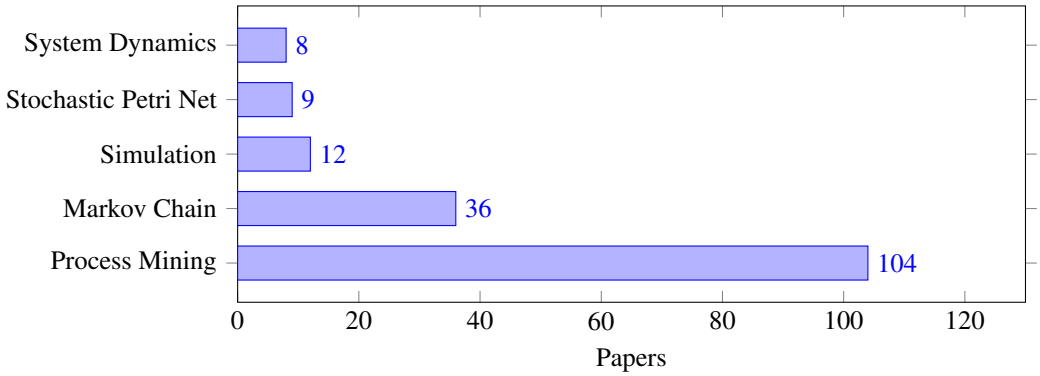


Figure 5: Process mining is the TF-PM family code on WS-151 (the 151 working-set studies that remain). The other four bars are tokens in `stochastic_technique`. They are not a partition of TF-ST, and a paper may carry more than one token.

for 110 rows, and 29 rows also contained `other`. Set beside TF-ST = 52, those two figures do not partition 169: 52 + 110 = 162 and 110 + 29 + 52 = 191. The four technique bars in Figure 5 are tokens in that column. They are not a partition of the 49.

RQ2.2: Predictive and stochastic techniques. The papers cited below illustrate Markov chains, stochastic Petri nets, and Monte Carlo simulation; they do not partition the 49 TF-ST studies on WS-151, and the System Dynamics token in Figure 5 is a fourth token in that column. The Markov-chain illustrations include models that represent software processes as discrete state transitions, including reinforcement learning agents grounded in Markov Decision Processes for pull-request outcome prediction [30], probabilistic risk quantification from repository data [31], defect removal effectiveness under code churn [32], agile state-space planning and tracking [33], release-time forecasting via generalised reliability models [34], and Stochastic Automata Networks for Waterfall team estimation [35]. The stochastic Petri net illustrations include models of concurrency and asynchrony in DevOps and CI/CD pipelines through non-Markovian specifications [36] and sensitivity analyses that identify bottleneck stages governing delivery probability [37, 38]. The Monte Carlo illustrations include project-level forecasting in hybrid System Dynamics models [39], agile risk assessment frameworks [40], cycle-time economic analysis [41], and the PRIMAD digital-twin architecture for SDLC [42].

Despite their methodological diversity, the cited descriptions share a parametrisation pattern. Transition probabilities, firing rates, and distributional parameters in these descriptions come from aggregate project statistics—historical defect counts, phase-level duration averages, or summary velocity metrics—rather than from transition frequencies extracted through process mining of fine-grained event logs [43, 37, 39]. In the same descriptions, model states stay anchored to coarse lifecycle phases. This is a reading of those descriptions. It is not a re-mark of the family codes labeled F3, which are the counts on WS-151 and the 293.

Three studies exemplify the productive convergence of process mining and stochastic modelling at the component level. Incerto [44] demonstrated that variable-length Markov chains outperform eighteen existing stochastic

conformance-checking techniques across ten of eleven benchmark datasets, establishing that higher-order sequential dependencies in event logs can be captured without sacrificing computational tractability. López-Pintado [45] extended process-simulation fidelity by replacing deterministic resource-availability calendars with probabilistic counterparts discovered directly from event logs, showing that stochastic resource representations more accurately replicate empirical cycle-time distributions. Jalote [46] took a complementary perspective, modelling programmer task sequences as first-order Markov chains derived from IDE interaction logs and demonstrating that transferring the process patterns of high-productivity developers to average-productivity peers yields measurable productivity gains. In the studies cited in this paragraph, the stochastic layer stays on one analytical concern.

Within the 380 studies of this corpus that carry a family mark, the work closest to a fully integrated multi-stage architecture is PRIMAD, proposed by Guinea-Cabrera [42], which orchestrates pm4py-based process discovery, Akka actor-system simulation, and LightGBM supervised learning within an evolutionary Digital Twin framework for the SDLC. Nevertheless, PRIMAD remains at what may be characterised as Level 2 integration: validation is reported per component rather than across the assembled pipeline, no formal inter-stage contracts govern the handoffs between discovery, simulation, and learning modules, and the architecture incorporates no dedicated Monte Carlo forecasting layer capable of propagating uncertainty estimates end-to-end. On the family codes of the 380, PRIMAD is the one study that matches the three families; none of the 74 added studies read from an abstract or a PDF adds a second triple. The scope of that count is stated in Section 4.1.

Studies that combine process mining with forecasting objectives (TF-PM ∩ TF-FC) tend to approach prediction through *process enhancement* rather than through explicit stochastic modelling. Gupta [47] demonstrated that combining inductive miner discovery with Markov chain-based predictive analytics enables identification of recurring failure patterns in software maintenance workflows. Pourbafrani et al. [48] presented SIMPT, a tool that extracts process trees with time parameters from event logs and enables interactive what-if simulation; its successor work [49] further integrates generative AI with business process simulation to produce conformance-preserving what-if scenarios. Caldeira et al. [50] mine IDE interaction logs and show that process complexity metrics derived from ProM correlate moderately ($r \approx 0.43$) with software cyclomatic complexity, achieving predictive accuracies exceeding 92%, demonstrating the feasibility of process-derived features for product-quality prediction.

Taken together, the papers cited in this paragraph describe PM output feeding a downstream model. None of these cited systems is described as reporting a probability distribution over completion time or remaining steps for an individual process instance.

4.7. RQ3: Integration and Gaps

RQ3 places each study on the L0–L3 scale (RQ3.1) and reads the gaps (RQ3.2) from the cells of that scale that are empty or nearly so.

RQ3.1: PM and stochastic integration level. The family codes on WS-151, which this revision did not re-mark, place rows on the L0–L3 scale defined in Table 15: the number of technique families TF-PM, TF-ST, and TF-FC a

study matches. The largest two-family cluster, comprising papers that touch TF-ST and TF-FC without TF-PM—including stochastic Petri net models of CI/CD pipelines [37, 38], Markov-based project estimation [35, 33], and Monte Carlo risk simulation [40, 41]—is coded **L1**, applying stochastic modelling to software processes without any process discovery layer. Papers spanning TF-PM and TF-ST, such as the VLMC conformance checker of [44] and the probabilistic resource-calendar discoverer of [45], are also coded **L1**: they couple discovered process structure with stochastic reasoning, but neither closes the loop toward distributional schedule forecasting. Papers combining TF-PM with TF-FC, such as [47] and [50], use process mining descriptively to assess workflows but lack the stochastic machinery required for Monte Carlo distributional forecasts. The PRIMAD framework [42] is the one triple match in these codes. It assembles event-log discovery, simulation, and ML prediction, and its evaluation is described as component-wise, so the code is L2 rather than a validated end-to-end contract. No row in the WS-151 codes is **L3**. L3 would be a row that matches the three families and states formal interfaces validated end-to-end on SDLC event logs.

The intersection counts on the same WS-151 codes are as follows. 1 of the 151 WS-151 papers (0.7%) jointly satisfies TF-PM, TF-ST, and TF-FC. Pairwise codes are TF-PM∩TF-ST in 8 papers (5.3%), TF-PM∩TF-FC in 5 papers (3.3%), and TF-ST∩TF-FC in 18 papers (11.9%). No row in the WS-151 codes is an **L3** architecture. The same triple on the 293 joint-table studies is also one paper. A rare code on these cuts is not a rare pattern in the literature. Figure 6 is the reading priority of the 73 PDFs. It is not the L0–L3 scale of Table 15.

RQ3.2: Gaps. Two readings of WS-151 bear on the gaps. First, the family-code count labeled F1: 104 papers (68.9%) match process mining and 27 (17.9%) match forecasting of a process metric. TF-PM with neither TF-ST nor TF-FC is 92 (60.9%). Second, the model assigned study types on these 151 rows: 66 (43.7%) case studies, 30 (19.9%) tool papers, 26 (17.2%) theoretical, 12 (7.9%) simulation-only, 8 (5.3%) controlled experiments, 5 (3.3%) surveys, and 4 (2.6%) residual categories (mixed, framework, exploratory; sum: 66 + 30 + 26 + 12 + 8 + 5 + 4 = 151). The public-dataset and replication-package percentages belong to the withdrawn F2 and are not reported; the reporting-quality scores in Table 8 do not stand in for them.

The gap these readings expose is one of integration rather than of coverage: process mining is present in two thirds of WS-151, stochastic modelling in a third, and the two meet in 8 studies. The evidence-type distribution in the 73-PDF subset (Figure 7) is a view of that subset, not a separate selection layer.

5. Discussion

5.1. Family-Code Counts F1–F5

The five items below collect the family-code counts of Section 4 under the labels used in the rest of the paper. F1, F3, and F4 are counts on the cut each sentence names (WS-151, 293, or the 380 for the triple); F2 and F5 are withdrawn. Their scope is the one stated in Section 4.1.

F1: Technique families on WS-151 and on 293. In the **working-set cut** (WS-151, 151 papers), 104 (68.9%) match TF-PM and 27 (17.9%) match TF-FC. TF-PM with neither TF-ST nor TF-FC is 92 (60.9%). In the **joint**

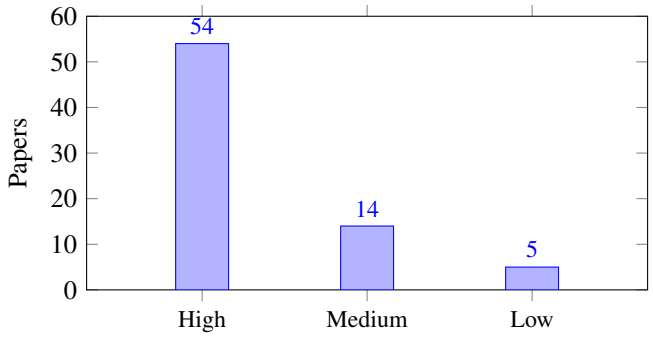


Figure 6: Reading priority of the 73 PDFs. High, Medium, and Low order those PDFs for reading. They are not the integration levels L0–L3 in Table 15 (122, 28, 1, and 0 on WS-151).

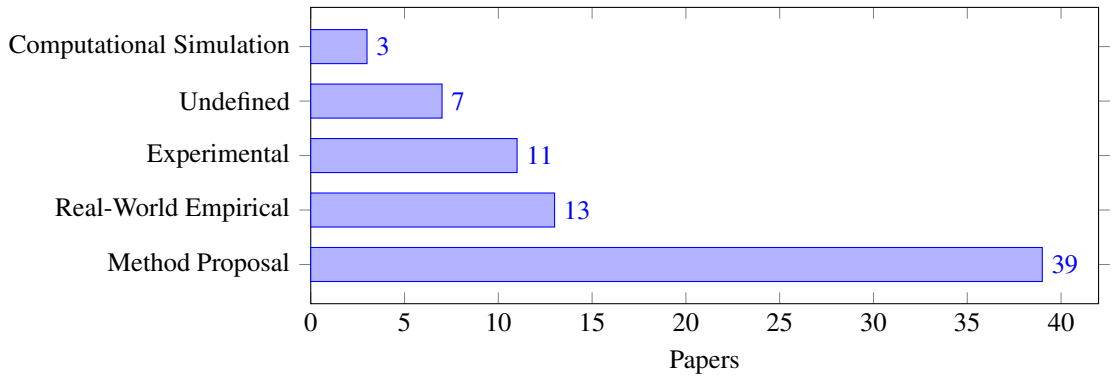


Figure 7: Evidence-type distribution in the 73-paper PDF triangulation subset.

cut (293 papers), 139 (47.4%) match TF-PM, 96 (32.8%) match TF-FC, and 123 (42.0%) match TF-PM with neither TF-ST nor TF-FC. The shift between the two cuts reflects the rows the auxiliary corpus added, which lean towards stochastic modelling without process mining.

F2: Event-log construction and data availability (withdrawn). These are free-text fields with 0% exact agreement against a human rater on the 31-study sample, and they were not re-extracted; no percentage is reported, and the reporting-quality scores do not replace them.

F3: Stochastic family on WS-151 and on 293. Working-set cut (WS-151): 49 of 151 papers (32.5%) match TF-ST, and the other 102 do not (49 + 102 = 151). TF-ST∩TF-FC covers 18 papers (11.9%). TF-PM∩TF-ST covers 8 (5.3%). Joint cut: 152 of 293 (51.9%) match TF-ST, and the other 141 do not (152 + 141 = 293). TF-ST∩TF-FC covers 80 (27.3%). TF-PM∩TF-ST covers 10 (3.4%). The joint cut contains more rows with TF-ST and without TF-PM than the working-set cut. The largest second-pass exclusion in the working-set queue is EC3 (232 of 886). The technique-name column (Figure 5) is not a partition of TF-ST and is not added to these family codes.

F4: Pairwise and triple intersections on WS-151, 293, and 380. The family marks added in this revision are the 74 studies read from an abstract or a PDF that meet a content criterion, inside the 380. None of those 74 has all three families, and no study in the 380 records a formal PM → Markov → MC → ML contract. Working-set cut (WS-151, 151 papers): the joint convergence TF-PM∩TF-ST∩TF-FC appears in 1 paper (0.7%). Pairwise convergences are TF-PM∩TF-ST: 8 (5.3%), TF-PM∩TF-FC: 5 (3.3%), TF-ST∩TF-FC: 18 (11.9%). Joint cut (293 papers): TF-PM∩TF-ST∩TF-FC remains 1 paper (0.3%), with pairwise convergences TF-PM∩TF-ST: 10 (3.4%), TF-PM∩TF-FC: 7 (2.4%), TF-ST∩TF-FC: 80 (27.3%). PRIMAD [42] is that one study inside the 380. It assembles discovery, simulation, and learning components but validates each independently and leaves the unified pipeline unevaluated. Its scope is SDLC-oriented: it builds a digital twin from issue-tracking flows in JIRA. It combines empirical process discovery (`pm4py`), a Python-based simulation core, an Akka actor system, and supervised machine learning for context-aware prediction. The reported results are component-wise: the paper does not validate a coupled PM → stochastic simulation → ML pipeline as one architecture.

F5: Process-derived features for prediction (withdrawn). Process-derived features and the coupling of a named mining algorithm with prediction are free-text fields with 0% exact match against a human rater on 31 studies; no percentage is reported, including the share of ML rows that name a process-model feature. The 19 IC4c title hits from the Scopus search of 31 October 2026 (Section 2.2) are a supplementary search result, not a substitute for this item.

5.2. Integration Levels and the SPMF Taxonomy

Table 15 defines the integration levels used throughout the article and reports their counts. A level is determined only by how many of the three *technique* families a study matches (TF-PM, TF-ST, TF-FC); TF-MSR describes the

data source and is not counted. With this single definition, the L1 and L2 rows coincide with the pairwise and triple intersections reported under F3 and F4.

Table 15: Integration levels of the WS-151 studies (the 151 working-set studies that remain). A level counts how many of TF-PM, TF-ST, and TF-FC the study matches.

Level	Operational definition	Papers	Example
L0	At most one technique family	122	Single-technique studies
L1	Exactly two technique families (TF-PM∩TF-ST: 7; TF-PM∩TF-FC: 4; TF-ST∩TF-FC: 17)	28	Incerto [44]; López-Pintado [45]
L2	All three technique families (TF-PM∩TF-ST∩TF-FC)	1	PRIMAD [42]
L3	All three families coupled through explicit interfaces and validated end-to-end	0	—
Total		**151**	

Note: L1 counts exclude the single L2 study, so they are one lower than the inclusive pairwise intersections in F3/F4 (8, 5, 18). The three exclusive pairs sum to 28 (7 + 4 + 17). Figure 6 reports reading priority for 73 PDFs. It is not this L0–L3 scale.

The authors' PATHCAST proposal is one candidate design for L3. It is **not** evaluated in this mapping study and is listed only as a research-agenda hypothesis (RA2–RA4, Section 5.3).

The integration levels order studies on one axis only. To position future work on more than that axis, the authors propose the Software Process Mining and Forecasting (SPMF) taxonomy (scope: Section 4.1), whose third dimension generalises the L0–L3 scale. Its three dimensions are:

Analytical Depth classifies studies along four levels (labelled AD to avoid confusion with the integration levels L0–L3): AD1 (Descriptive: discovery, conformance), AD2 (Diagnostic: bottleneck, rework), AD3 (Predictive: outcome, remaining time), and AD4 (Forecasting: distributional, probabilistic).

SDLC Coverage maps which lifecycle phases are addressed: Requirements and Design, Development (coding, commits, PRs), Quality (testing, reviews, bugs), Delivery (CI/CD, release), and Operations (incidents, maintenance).

Technique Integration classifies methodological combination: Single (PM only, ML only, MC only), Dual (PM + ML, PM + Markov, Markov + MC), and Pipeline (PM → Markov → MC → ML).

5.3. Research Agenda and Implications for Practitioners

The four directions below are proposals by the authors, grounded in their reading of the studies cited in Section 4; the family-code counts F1–F5 situate each direction but do not derive it (Section 4.1).

RA1: Standardized Event Log Construction for SDLC. Target: frameworks for transforming heterogeneous repository data into comparable event logs with cross-tool case correlation. Related count: F2 (withdrawn), which is why the need is argued from the cited studies in Section 4.5 rather than from a percentage.

RA2: Process-Aware Stochastic Forecasting. Target: formal integration of mined transition structure with Markov and Monte Carlo forecasting. Related count: F3, where the TF-PM∩TF-ST intersection is 8 of 151 on WS-151 and 10 of 293 on the joint cut.

RA3: Process-Derived Features for ML. Target: systematic use of conformance, rework, and path-entropy features in SE predictors. Related count: F5 (withdrawn).

RA4: Empirical Benchmarking of Integrated Pipelines. Target: comparable OSS benchmarks with temporal hold-out for end-to-end PM–stochastic–ML pipelines. Related count: F4 (one triple, PRIMAD, validated per component). Formal specification and evaluation of candidates such as PATHCAST remain future work by the authors.

For practitioners, the mapping supports two observations, both drawn from the cited studies rather than from the family-code counts. First, the techniques that forecast SDLC metrics from event data in the corpus (Section 4.6) operate on logs that already exist in issue trackers and version control, so the entry cost is in event-log construction (RA1) rather than in data collection. Second, no study in the 380 reports an end-to-end PM–stochastic–ML pipeline validated as one architecture, so a practitioner adopting such a pipeline today is adopting an untested combination of individually tested parts.

6. Threats to Validity

6.1. Internal Validity

LLM-based screening introduces the risk of systematic decision bias. We report six aspects of how this risk was handled: (i) structured outputs; (ii) author inspection of `include` decisions; (iii) human validation of the selection against a two-rater reference standard; (iv) LLM–LLM cross-model agreement; (v) the T/A false-positive rate; and (vi) the closure rule for unresolved cases used in the original run.

(i) **Structured outputs**. All screening responses use a strict JSON schema, enabling post-hoc analysis of decision patterns and rationale distributions.

(ii) **Author inspection of LLM `include` decisions**. Every paper retained at the FT stage of the original run was inspected by the first author against the PRISMA flow before being added to the 169-study working-set tier. The escalation set was judged by both reviewers (Section 6).

(iii) **Human validation of the selection**. *T/A stage.* A stratified random 20% sample of working-set T/A decisions (n = 468) was screened by a human rater blind to the LLM. In the working set, LLM `maybe` decisions were forwarded

to FT screening (Section 3); against the human labels, the operational T/A Recall is therefore 72/72 = 1.000 (Wilson 95% CI 0.949–1.000). Counting `maybe` as a negative, as the R1 manuscript did, gives 18/72 = 0.250, but that figure describes a decision rule the working set never applied.

FT stage — R1 reference standard withdrawn. The FT reference standard reported in R1 (n = 177) did not record eligibility: all 126 human `include` labels coincided with papers for which a PDF had been obtained, and 50 of the 51 human `exclude` labels were annotated "PDF not accessible". The resulting Recall (0.246) measured PDF availability, not screening error, and is withdrawn.

Second pass — round-2 validation. The replication files label this pass FT. Before any new judgement we pre-registered the check (`PREREGISTRATION_round2_validation.md`; success if the point estimate of Recall is at least 0.90). The same 177 records were judged against Table 6 by two raters, independently, blind to the LLM and to each other. Disagreements were resolved by consensus. A record whose text could not be obtained is `not_assessable` and leaves the denominator. Six records are in that class. Agreement between the raters, before consensus, is $\kappa = 0.654$.

The reference standard is the consensus after a full-text re-reading of the 23 records the first consensus had included and the LLM had excluded. Five of those 23 are ineligible. One is not assessable, because the file on disk is a conference programme. Seventeen remain eligible. Against that reference the LLM's Recall on the working-set sample is 0.630 (29 true positives out of 46 eligible studies; Wilson 95% CI 0.486–0.755). This is not a full-text Recall. A PDF was read for 0.5% of the 886 records, and the second-pass prompt received the title and the abstract. Three further figures are reported on the confusion matrix of Table 16 (TP = 29, FP = 5, FN = 17, TN = 120), as fixed in the pre-registration. Lost evidence is the share of eligible sampled studies the screen excluded, $FN/(TP + FN) = 1 - \text{Recall} = 0.370$. The Matthews correlation coefficient [51, 52],

$$\text{MCC} = \frac{TP\,TN - FP\,FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}},$$

is 0.656. WMCC is a cost-weighted variant of the same statistic in which each false negative is counted ten times before the formula is applied, $\text{WMCC} = \text{MCC}(TP, FP, 10\,FN, TN)$; the ten-to-one ratio encodes the judgement that a missed eligible study costs a review more than a wrongly forwarded one, in line with the recall-first stance of LLM4SCREENLIT [18]. On this matrix WMCC is 0.168. Both coefficients are computed by the script `human_kappa.py` in the replication package. The same Recall, 0.630, is obtained if the second rater's labels are used alone. The first author had computed the round-1 agreement report on this sample and may remember some LLM decisions. The second rater had no such exposure. The result is below 0.90, so the pre-registered rule required a human re-screen of the accessible working-set records the LLM had excluded.

Auxiliary tier. The same rule required a check of LLM non-includes outside the working set. The 3,807 auxiliary records partition into the 235 LLM includes (S0) and three strata of non-includes: S1, title/abstract excludes (N_1 = 2,423); S2, second-pass excludes, including the re-screen after abstract enrichment (N_2 = 537 = 291 + 246); and S3, records closed without a decision under the extended exclusion rule (N_3 = 611 = 595 + 16). The strata sum to 3,806: one record (`d428a409`) has no stored title/abstract decision in the auxiliary screening file, which is why N_1 is 2,423 rather than the 3,807 − 1,383 = 2,424 implied by the forwarded count; that record was excluded under EC1 and

EC3 in the 200-paper sensitivity sample and is in no stratum and not in the corpus. Random samples of 40, 40, and 60 records were drawn (seed 20261006). The first author judged each sampled record. The second author judged a 30% subset ($\kappa = 0.494$). Twenty auxiliary includes were mixed into the sheets so the rater could not infer that every row was an exclusion. Records whose eligibility could not be judged were labelled `not_assessable` and dropped from the denominator, as the pre-registration specified: 0 in S1, 1 in S2, and 9 in S3, leaving $n_1 = 40$, $n_2 = 39$, and $n_3 = 51$ assessable records, of which $k_1 = 1$, $k_2 = 8$, and $k_3 = 1$ were judged `include`. The pre-registered estimator of missed studies is $\hat{M} = \sum_h N_h\, k_h/n_h$: $2{,}423 \times 1/40 = 60.6$, $537 \times 8/39 = 110.2$, and $611 \times 1/51 = 12.0$ (Wilson 95% intervals for k_h/n_h: 0.004–0.129, 0.108–0.355, 0.003–0.103), summing to 182.7 before rounding. The sample estimator of auxiliary Recall is $235/(235 + 182.7) = 0.563$, which is what triggered the full re-screen.

Both reviewers then screened the working-set exclusions and the rest of S1, S2, and S3, including the records whose first label was heuristic. They agreed with 92% of those suggested labels. The sheets (4,005 rows) store a decision equal to the suggestion on every row, so the labels they did not accept are not marked there. Rebuilt from the title or abstract the LLM treated as sufficient, that share of the coded includes is the 32 studies in `labels_ta_insufficient_for_include.csv` (32 of 338, 9.5%), and those 32 are removed from the corpus. The counts of missed studies below follow the suggested includes. The suggested includes contain 89 studies the LLM had missed (10 in the sample, 33 in the rest of S1, 32 in the rest of S2, 14 in the rest of S3). Auxiliary Recall on that count is 0.725 (235 of 324). It is still below 0.90. Seventeen of the 33 S1 includes, and all 14 S3 includes, began as heuristic suggestions. The reviewers' 92% agreement covers those suggestions. It does not replace them with a second written code on each row, so the 150 added studies rest on suggested labels accepted by two reviewers and none received a second independent coding. After de-duplication, 168 records were new to the list. The eligibility pass removes 18 of them, so the studies new to the corpus are ADD-150 in Section 4. The 89 is the auxiliary slice of the screen. ADD-150 also includes working-set records the LLM had excluded.

Table 16: Round-2 check of the working-set sample. The human label is the consensus reference. Lost evidence is 1 − Recall. The interval is a Wilson 95% confidence interval. Six records judged `not_assessable` are outside this table. The Recall in this table is not a full-text Recall: a PDF was read for 0.5% of the 886 records, and the prompt received the title and the abstract.

	TP	FP	FN	TN	Recall	95% CI	Lost ev.
Working-set sample	29	5	17	120	0.630	(0.486, 0.755)	0.370

The two tables that follow are the round-1 matrices. The row labelled FT (withdrawn) measured whether a PDF was on disk (Recall 0.246; $\kappa = 0.122$). That row is withdrawn. The 0.630 above is the Recall of the working-set sample. It is not a full-text Recall.

QA and extraction. On the 31 papers included by both the R1 human rater and the LLM, LLM QA matched the human rater on 68–100% per criterion; free-text extraction fields showed 0% exact-match agreement. Those fields were not re-extracted by a human. F2 and F5, which use them, are withdrawn.

Table 17: Round-1 agreement (Cohen's κ) between the primary LLM screener (claude-haiku-4-5) and an independent human rater on a stratified random 20% sample. Interpretation thresholds follow Landis and Koch (1977) [24]. The T/A row is the title/abstract check. The row "FT (withdrawn)" ($\kappa = 0.122$, $n = 177$) uses the same PDF-on-disk labels as the withdrawn Recall 0.246. It is not agreement on the criteria, and it is not the κ of this revision.

Stage	N	κ_{multi} (interpretation)	κ_{binary} (interpretation)
T/A	468	0.250 (fair)	0.335 (fair)
FT (withdrawn)	177	0.122 (slight)	0.122 (slight)

Table 18: Round-1 human-vs-LLM confusion on a stratified random 20% sample (human rater as gold standard; counts, not percentages). Recall and Lost Evidence (1−Recall) use Wilson 95% confidence intervals. The T/A row labelled "maybe→positive" treats LLM `maybe` decisions as positive referrals, matching LLM4SCREENLIT R6. The row "FT (withdrawn)" is not a screening result: Recall 0.246 records whether a PDF was on disk. The Recall of this revision is 0.630 on the working-set sample (Table 16). That 0.630 is not a full-text Recall.

Stage	TP	FP	FN	TN	Recall	95% CI	Lost Ev.
T/A (strict)	18	4	54	392	0.250	(0.164, 0.361)	0.750
T/A (maybe→pos.)	72	105	0	291	1.000	(0.949, 1.000)	0.000
FT (withdrawn)	31	3	95	48	0.246	(0.179, 0.328)	0.754

(iv) **Cross-model inter-rater agreement (Cohen's κ).** A separate stratified random 20% sample of T/A ($n_{\text{theoretical}} = 468$ from the 2,340-paper working set) and FT ($n_{\text{theoretical}} = 177$ from the 886-paper FT queue) decisions was independently re-rated by a second model from the same provider (*claude-sonnet-4-6*). Of the 468 T/A papers in the sample, 420 produced a parseable decision from the verifier and entered the agreement computation; the remaining 48 papers returned a non-parseable response because the abstract was absent, and the verifier then refused a categorical answer. Those 48 are part of the verifier workload: 48 of 468 sampled title/abstract papers. They stay out of κ because the verifier returned no category. On the 420 parseable papers the binary observed agreement is 410 of 420. If each of the 48 is scored as a disagreement, that observed agreement is 410 of 468. They are not added to Lost Evidence. Lost Evidence is the human false-negative rate on the pre-registered sample, and these 48 are verifier responses with no category. The reported κ is the agreement on the 420 papers that produced a parseable decision, out of 468 sampled. All 177 FT sample papers produced a parseable decision. Both the multi-class κ (over the original `include`/`exclude`/`maybe` or `include`/`exclude`/`pending` categories) and the binary κ (collapsing the rater outputs to `include` versus `not-include`) are reported in Table 19. The binary κ is reported as a secondary signal: both binary κ values (0.695 for T/A and 0.694 for FT) exceed the substantial-agreement threshold $\kappa \geq 0.61$ [24], but two LLMs from the same provider may share systematic biases that inflate LLM–LLM agreement relative to an independent human check. The multi-class FT κ is lower (0.479, moderate) because the verifier returned `pending` on 25/177 FT papers where the primary screener committed to a binary decision.

Table 19: Inter-rater agreement (Cohen's κ) between primary screener (claude-haiku-4-5) and verifier (claude-sonnet-4-6) on a stratified random 20% sample. Interpretation thresholds follow Landis and Koch (1977) [24].

Stage	N	P_o	P_e	κ_{multi} (interpretation)	κ_{binary} (interpretation)
TA	420	82.1%	50.2%	0.641 (substantial)	0.695 (substantial)
FT	177	76.3%	54.5%	0.479 (moderate)	0.694 (substantial)

The cross-model verification protocol was also extended to the auxiliary tier (Table 20). At the second pass the sample is a stratified random 20% of the 1,383 forwarded records (n = 276). On it the verifier (Sonnet 4.6) returned `pending` on 40 of the 42 papers the primary screener (Haiku 4.5) had rated `include` and `exclude` on the other 2; it confirmed none. The resulting multi-class κ = 0.004 and binary κ = 0.000 (Po = 84.8%) therefore mean that the cross-model check provides no confirmation of the 212 auxiliary includes, which is a different situation from the T/A row and the reason the auxiliary includes were afterwards read by the human raters. The T/A row is not a 20% sample: of the 761 records drawn (20% of 3,806 with a decision), only 39 received a verifier decision, all of them primary `excludes`, of which the verifier confirmed 38; there Po = 97.4% with κ = 0.000 is a base-rate artefact of a one-class slice, not evidence about agreement. The auxiliary records were afterwards screened by both reviewers, who agreed with 92% of the suggested labels (Section 6). The combined-tier counts rest on the family codes of the original coded set, plus the triple-intersection count on the 380 (Section 4.1).

Table 20: Auxiliary-tier cross-model agreement (Haiku 4.5 primary vs. Sonnet 4.6 verifier). FT: stratified 20% sample of the 1,383 forwarded records. TA: only 39 of the 761 records drawn (20% of 3,806) received a verifier decision, all primary `excludes`; that row is not a 20% sample.

Stage	N	P_o^{multi}	P_o^{binary}	κ_{multi}	κ_{binary}
TA	39	97.4%	97.4%	0.000 (slight)	0.000 (slight)
FT	276	54.3%	84.8%	0.004 (slight)	0.000 (slight)

(v) **T/A false-positive rate**. The LLM confirmatory pass at FT level (Round 1) rejected 33 of the 111 T/A `include` papers (29.7%; Table 12), primarily for EC3 (algorithm without SE evaluation). This false-positive rate is consistent with the T/A-screening false-positive rates reported by Khraisha et al. [23] for medical SLRs (in the 25–35% range, depending on domain specificity) and by Syriani et al. [22] for SE SLRs. A false positive at title/abstract can be removed at the second pass. A false negative is not removed by that pass. Recall on the working-set sample is 0.630. That figure is not a full-text Recall: a PDF was read for 0.5% of the 886 records. Missed studies would affect any claim drawn from the coded rows. This revision draws none from F1–F5. The Round 2 screening reprocessed 495 `maybe` papers with enriched abstracts and brought the working-set tier to 169. The corpus of this revision is the 456 in Table 14.

(vi) **Original closure of unresolved uncertainty (revised in resubmission)**. In the original execution run, papers

where metadata remained insufficient or verifier disagreement could not be resolved were closed under EC5-extended, biasing toward false negatives. In the working set this rule affected only records that remained abstract-insufficient at FT; in the auxiliary tier it closed 611 records (stratum S3), whose share of relevant studies is estimated in item (iii). The revised protocol routes `maybe`/`pending` to human review instead (Section 2.4).

6.2. *External Validity*

The search strings may exclude studies that use non-canonical terminology. This threat is mitigated through three mechanisms: the complementary MSR string, the frontier stochastic-PM string, and one snowballing iteration over 595 records attributed to 400 source DOIs (Section 2.2). The ten control papers were recovered, but this is a weak check: they were known when the strings were written, so their recovery does not bound the proportion of relevant studies the strings miss. Those ten papers are not the snowballing seed list. Beyond terminology, the review is limited to English-language peer-reviewed publications, which leaves out work in other languages. The search window is January 1994 to 12 April 2026, over five databases. The window states the period searched; it does not remove the language restriction, and the search does not cover grey literature, which is excluded. The stop rule is the document type at search time: journal articles, conference papers, and workshop papers. Theses, preprints, and technical reports were not searched in a separate round. The auxiliary identifier list (3,807 rows; Section 3) was screened in full. It shares 364 papers with the working set. The mapped set is the 456 in Table 14. The working set was built from Scopus, IEEE, and ACM. SpringerLink, Web of Science, and most of the snowballing records were screened in the auxiliary tier, with greater use of abstracts. A difference between a percentage on 169 studies and a percentage on 318 studies can come from that unequal screen.

A four-stage sensitivity check was executed against the auxiliary corpus. First, a random 200-paper subsample was re-screened with the same T/A LLM protocol used for the working set; 12 of 200 (6.0%) returned a T/A `include` decision and 59 (29.5%) returned `maybe`, yielding a point-estimated 228 additional candidate includes if the auxiliary corpus were screened in full. Given that this estimate exceeded the working-set include count, the second stage was triggered: a screen of all 3,807 auxiliary papers at title/abstract and at the second pass. The result is reported in Table 22: of the 3,807 auxiliary papers, 232 were T/A-included and 1,151 T/A-marked-maybe, of which 212 survived the second pass as `include`. The arithmetic from there to the 456 is the one in Table 14.

In the original run, 16 auxiliary records stayed pending after abstract enrichment (EC5c), and 595 records still had no abstract (EC5b). Those records were closed under EC5. In this revision the two reviewers screened the pending and the unclassifiable records (Section 6).

Table 23 reports the re-screen of the 285 auxiliary records that gained an abstract through enrichment. A second external-validity exposure concerns the original 148-paper EC5 stratum (full text inaccessible at first pass). To bound the residual recall risk, an automated open-access (OA) recovery pass was executed against four public discovery services (Unpaywall, Semantic Scholar Graph API, OpenAlex, CORE). The result is summarised in Table 24: 27 of the 148 EC5 papers (18.2%) returned at least one OA URL during the recovery pass; of those, 17 PDFs were

Table 21: Auxiliary-corpus sensitivity check. A random sample of n papers from the deferred auxiliary corpus was re-screened with the same T/A LLM protocol used for the working set.

Quantity	Value
Sample size	200 (seed=20260428)
Auxiliary corpus total	3807
Working-set confirmed includes	169
Sample T/A include rate	6.0% (n_{inc} = 12)
Sample T/A maybe rate	29.5% (n_{maybe} = 59)
Sample T/A exclude rate	64.5% (n_{exc} = 129)
Expected new includes (point estimate)	228
Expected new includes (upper bound, maybe→include)	1351
Sensitivity ratio versus the 169 working-set includes (point)	135.2%
Sensitivity ratio versus the 169 working-set includes (upper)	799.7%

Table 22: Auxiliary-corpus FT-screening outcome (papers that passed T/A on the 3,807-paper deferred corpus).

Outcome	**Count**	**%**
Forwarded from T/A	1383	100.0
Included (new)	212	15.3
Excluded	291	21.0
Pending / unparsed	880	63.6
Combined set, first auxiliary pass (raw, pre-deduplication)	**381**	—

Table 23: Re-FT screening of aux-pending papers that gained an abstract through the post-hoc enrichment cascade.

Outcome	**Count**	**%**
Re-FT include	23	8.1
Re-FT exclude	246	86.3
Re-FT pending	16	5.6
Total re-FT	**285**	100.0

successfully downloaded (11.5% of the stratum). A targeted audit of those 17 files found that 14 OA links resolved to unrelated papers or generic proceedings and therefore remained under EC5, while 3 papers were substantively re-screened and reclassified to EC3. The active PRISMA EC5 count after this audit is thus 145 rather than 148, decomposed as 121 + 10 + 14: 121 papers (81.8% of 148) for which no OA URL was found, 10 for which a URL was found but no PDF could be downloaded, and 14 whose downloaded file was an unrelated document. All 145 continue to be reported under EC5 in the PRISMA flow. The same OA-download pass was run over the 212 auxiliary includes and recovered 32 PDFs (15.1% of 212; the lower yield reflects the publisher-paywall mix of the auxiliary corpus); across its 239 targets (212 auxiliary includes plus the 27 EC5 records with an OA URL) the pass downloaded 32 + 17 = 49 files (20.5%).

Table 24: EC5 recovery attempt against four open-access discovery APIs.

Source	Recovered	% of EC5
Unpaywall	6	4.1%
Semantic_Scholar	6	4.1%
Openalex	6	4.1%
Core	20	13.5%
Any source	**27**	**18.2%**

6.3. Conclusion Validity

F1, F3, and F4 are the original family codes on the cut named in the sentence (Section 4.1); the human re-mark in this revision is the triple intersection on the 74 added studies with an abstract or a PDF, none of which joins the three families, together with the third-family audit of the 99 two-family studies (PDF for 78, abstract for 21; perfect agreement on the 30 double-read rows; `results/family_code_audit/`), which moved none from L1 to L2 and found none at L3. The set on which a single coding error would change the conclusion was those 99 (98 L1 studies and PRIMAD), 29 of them in WS-151. F2 and F5 are withdrawn: on 31 studies the exact agreement between the model and a human rater on those free-text fields is 0%, and agreement on the categorical fields is 29–81%, without adjudication. The two extraction layers are (i) the 169-study working-set extraction (Section 2.6; Table 9), partly checked against the 73-PDF subset, and (ii) the 212-study auxiliary extraction from abstracts. The quality scores are what the same model assigned. A score taken from an abstract is not a quality assessment of the study. The 4/8 line is not a filter on any statement. Of the WS-151 studies, 124 were assigned at least 4/8 (mean 4.90 ± 1.36). Of the 212 auxiliary studies, 179 were (mean 5.09 ± 1.43). Of the 293 joint-table studies that remain, 241 were (82.3%, mean 4.96 ± 1.40). Abstract-only scores stay labeled as abstract-only scores. Table 25 is the earlier file: 169 working-set rows and 212 auxiliary rows, 381 before the duplicate removal that gave the 318. Its 136 of 169 is not the 124 of WS-151 above, and its 315 of 381 is not the 241 of 293. The two tables are different cuts, not the same result.

Table 25: Model-assigned quality scores on the 169 working-set studies and the 212 auxiliary studies (n = 381, before the duplicate removal that yields the 318 joint tables). A score of 4/8 is not a filter.

Tier	N	Mean	SD	$\geq 4/8$	$< 4/8$
Working-set	169	4.88	1.40	136	33
Auxiliary	212	5.09	1.43	179	33
Combined	**381**	**5.00**	**1.42**	**315**	**66**

A residual conclusion-validity threat is the LLM-based extraction of PM-technique, stochastic-technique, and SDLC-phase fields. To mitigate this, the 73-PDF triangulation subset was used to spot-check the extracted fields against the corresponding full texts; no systematic disagreement was observed in the spot-check, but human re-extraction on the stratified sample (n = 31 papers included by both human and LLM) shows 0% exact-match agreement on free-text fields and 29–81% on categorical fields (Section 6, item iii).

6.4. Reproducibility and Use of AI

In line with current Elsevier and Springer-Nature editorial policy, this review explicitly declares the use of AI tools in its conduct. The primary screener at title and abstract, including the second pass, is *claude-haiku-4-5-20251001* (Anthropic), invoked through the Anthropic Message Batches API for screening and through the synchronous Messages API for QA scoring. Those calls omit `temperature`; the API default is 1. The cross-model inter-rater verification described above uses *claude-sonnet-4-6* (Anthropic) as the second rater. AI tools are **not** listed as authors of this work and are responsible for none of the editorial judgements; final selection, the decision not to filter on a quality score, taxonomy design, and the wording of the synthesis are the responsibility of the authors.

A complete replication package is deposited at Zenodo (DOI 10.5281/zenodo.21939600). It contains the search strings for all five databases; deduplication and enrichment scripts; screening and QA prompts with JSON schemas; per-stage decision tables; reporting-quality and extraction tables for both corpus tiers; LLM–LLM and human–LLM agreement reports; the 456-study list of this revision (`included_studies_506.csv`, 456 rows); a completed SEGRESS checklist; and a PRISMA flow snapshot. File-level layout is documented in the package README. The deposit is intended to be self-contained: a reviewer can re-run the LLM-assisted protocol on the deposited working set under the same model identifiers.

7. Conclusion

This mapping study makes two contributions. The first is methodological: a pre-registered, two-rater measurement of the selection error of an LLM-assisted screen on a software engineering corpus. Against a success bar of Recall ≥ 0.90, the screen reached 0.630 on the working-set sample (29 of 46; Wilson 95% CI 0.486–0.755) and 0.725 on the

auxiliary tier (235 of 324). The shortfall has three named mechanisms: a second pass that read the abstract rather than the PDF for 99.5% of the 886 records, 595 records without an abstract closed by rule, and 32 of 338 coded includes (9.5%) whose title or abstract does not support inclusion. Operational title/abstract Recall with `maybe` counted as a referral is 1.000 (72 of 72), which locates the working-set loss at the abstract-based second pass rather than at triage. The human re-screen that the pre-registered rule triggered recovered 150 studies.

The second contribution is the corpus itself: 456 publications at the intersection of process mining, stochastic modelling, and forecasting of software processes (Table 14), 306 that remain from the coded list of 338 plus the 150 added studies (ADD-150), classified into the four families TF-PM, TF-ST, TF-FC, and TF-MSR wherever an abstract or a PDF was available (380 of the 456), summarised in Appendix B and listed in Supplementary Material S1. On WS-151, the 151 working-set studies with full codes, 104 carry process mining, 49 a stochastic model, and 27 forecasting; the two technique families meet in 8 studies; and on the 380 studies with at least one family mark, one study, PRIMAD, joins all three, with its components validated separately; after re-examining every two-family study for a missing third family (n = 99), none reaches L3 as defined in Table 15, the level PATHCAST targets, so within the reviewed corpus that architecture has no published equivalent; PATHCAST itself is not evaluated here. Those counts describe the named cuts of this corpus (Section 4.1); the F2 and F5 fields are withdrawn after the extraction check. A supplementary Scopus search on 31 October 2026 with machine-learning terms found 19 further IC4c titles, none of which joins the three families.

On that basis the authors propose the SPMF taxonomy and the agenda RA1–RA4 (Section 5.3); the PATHCAST pipeline is one candidate for RA4 and is future work. The replication package, with the 456-study list, the screening prompts, and the human-check sheets, is openly available at Zenodo (DOI 10.5281/zenodo.21939600).

Appendix A. Search strings as executed

This appendix reproduces, verbatim, the database-specific strings behind the rows of Table 3. Section 2.2 gives the shared term lists from which they were derived.

`scopus_principal` **(1,595 records)**

```
TITLE-ABS-KEY( ("software engineering" OR "software development" OR
"software process" OR "software lifecycle" OR "SDLC" OR "DevOps" OR "CI/CD" OR "continuous integration"
OR "agile development" OR "issue tracking" OR "version control" OR "code review" OR "pull request" OR
"software repository" OR "software evolution") AND ("process mining" OR "process discovery" OR
"conformance checking" OR "workflow mining" OR "predictive process monitoring" OR "event log analysis"
OR "process intelligence" OR ("Monte Carlo simulation" AND ("software" OR "process mining" OR
"throughput" OR "lead time")) OR ("Markov chain" AND ("software" OR "process mining" OR "workflow" OR
"development process")) OR (("stochastic model" OR "transition probability" OR "absorbing chain") AND
("software" OR "process mining")) OR (("remaining time prediction" OR "outcome prediction" OR "cycle
time prediction" OR "lead time prediction") AND "software") OR ("event log" AND ("software" OR
"repository" OR "JIRA" OR "GitHub"))) ) AND PUBYEAR > 1993 AND PUBYEAR < 2027 AND (LIMIT-TO(DOCTYPE,
"ar") OR LIMIT-TO(DOCTYPE, "cp")) AND LIMIT-TO(LANGUAGE, "English")
```

scopus_markov_testing **(498 records)** TITLE-ABS-KEY((("Markov chain" OR "Markov model") AND ("statistical software testing" OR "software testing" OR "software reliability")) OR ("A Markov chain model for statistical software testing")) AND PUBYEAR > 1993 AND PUBYEAR < 2027 AND (LIMIT-TO(DOCTYPE, "ar") OR LIMIT-TO(DOCTYPE, "cp")) AND LIMIT-TO(LANGUAGE, "English")

scopus_stochastic_petri **(6 records)** TITLE-ABS-KEY((("stochastic Petri net" OR "stochastic workflow net" OR "GSPN") AND ("remaining service time" OR "service execution time" OR "remaining time")) OR ("Prediction of Remaining Service Execution Time Using Stochastic Petri Nets with Arbitrary Firing Delays")) AND PUBYEAR > 1993 AND PUBYEAR < 2027 AND (LIMIT-TO(DOCTYPE, "ar") OR LIMIT-TO(DOCTYPE, "cp")) AND LIMIT-TO(LANGUAGE, "English")

scopus_simulation_bridge **(200 records)** TITLE-ABS-KEY((("process mining" OR "process discovery" OR "workflow mining") AND ("simulation model" OR "simulation models" OR "process simulation" OR "discovering simulation models" OR "colored Petri") AND ("event log" OR "process model" OR "business process")) OR ("Discovering Simulation Models")) AND PUBYEAR > 1993 AND PUBYEAR < 2027 AND (LIMIT-TO(DOCTYPE, "ar") OR LIMIT-TO(DOCTYPE, "cp")) AND LIMIT-TO(LANGUAGE, "English")

scopus_complementar **(48 records)** TITLE-ABS-KEY(("mining software repositories" OR "software repository mining" OR "software archaeology" OR "mining commit logs" OR "GitHub mining" OR "Jira mining") AND ("process model" OR "workflow" OR "state transition" OR "Markov" OR "Monte Carlo" OR "stochastic" OR "lead time" OR "cycle time" OR "throughput" OR "transition matrix" OR "sojourn time")) AND PUBYEAR > 1993 AND PUBYEAR < 2027 AND (LIMIT-TO(DOCTYPE, "ar") OR LIMIT-TO(DOCTYPE, "cp")) AND LIMIT-TO(LANGUAGE, "English")

scopus_fronteira **(30 records)** TITLE-ABS-KEY(("Monte Carlo" OR "Markov chain" OR "stochastic simulation" OR "probabilistic forecast" OR "stochastic model" OR "absorbing Markov" OR "transition matrix") AND ("process mining" OR "event log" OR "workflow mining" OR "process model" OR "business process") AND ("software development" OR "software engineering" OR "software process" OR "DevOps" OR "SDLC" OR "issue tracker" OR "commit log" OR "build pipeline")) AND PUBYEAR > 1993 AND PUBYEAR < 2027 AND (LIMIT-TO(DOCTYPE, "ar") OR LIMIT-TO(DOCTYPE, "cp")) AND LIMIT-TO(LANGUAGE, "English")

wos_principal **(632 records)** TS=("process mining" OR "process discovery" OR "conformance checking" OR "workflow mining" OR "event log analysis" OR "process intelligence" OR "predictive process monitoring") AND TS=("software engineering" OR "software development" OR "software process" OR "SDLC" OR "DevOps" OR "continuous integration" OR "agile development" OR "issue tracking" OR "code review" OR "software project" OR "software repository" OR "software maintenance" OR "software evolution")

wos_complementar **(338 records)** TS=(("Monte Carlo simulation" OR "Markov chain" OR "stochastic model" OR "transition probability" OR "absorbing chain" OR "process simulation") AND ("software" OR "process mining" OR "workflow" OR "throughput")) AND TS=("software engineering" OR "software development" OR "software process" OR "SDLC" OR "DevOps")

wos_msr **(26 records)** TS=("mining software repositories" OR "software repository mining" OR "software archaeology" OR "mining commit logs" OR "GitHub mining" OR "Jira mining") AND TS=("process model" OR "workflow" OR "state transition" OR "Markov" OR "Monte Carlo" OR "stochastic" OR "lead time" OR "cycle time" OR "throughput" OR "transition matrix" OR "sojourn time")

`wos_fronteira` **(1 record)** `TS=("Monte Carlo" OR "Markov chain" OR "stochastic simulation" OR "probabilistic forecast" OR "stochastic model" OR "absorbing Markov" OR "transition matrix") AND TS=("process mining" OR "event log" OR "workflow mining" OR "process model" OR "business process") AND TS=("software development" OR "software engineering" OR "software process" OR "DevOps" OR "SDLC" OR "issue tracker" OR "commit log" OR "build pipeline")`

`acm_principal` **(24 records)** `(Abstract:"process mining" OR Abstract:"conformance checking" OR Abstract:"process discovery" OR Abstract:"workflow mining" OR Abstract:"event log analysis" OR Abstract:"process intelligence" OR Abstract:"predictive process monitoring") AND (Abstract:"software engineering" OR Abstract:"software development" OR Abstract:"software process" OR Abstract:"SDLC" OR Abstract:"DevOps" OR Abstract:"continuous integration" OR Abstract:"agile development" OR Abstract:"issue tracking" OR Abstract:"software repository" OR Abstract:"software project")`

`acm_msr` **(32 records)** `(Abstract:"mining software repositories" OR Abstract:"software repository mining" OR Abstract:"software archaeology" OR Abstract:"GitHub mining" OR Abstract:"Jira mining") AND (Abstract:"process model" OR Abstract:"workflow" OR Abstract:"state transition" OR Abstract:"Markov" OR Abstract:"Monte Carlo" OR Abstract:"lead time" OR Abstract:"cycle time" OR Abstract:"throughput" OR Abstract:"process mining" OR Abstract:"process analysis")`

The third ACM export, `acm_extra_refs` (40 records), is a supplementary BibTEX file and has no query string. The four IEEE protocol strings follow. The raw file records 17 manual batches and 1,386 records. It does not record which of these four strings produced each batch. `acm_complementar` and `springer_11`–`springer_13` are in `config/queries.py` and were not executed.

`ieee_exec1` `("process mining" OR "conformance checking" OR "process discovery" OR "workflow mining" OR "event log analysis" OR "predictive process monitoring") AND ("software engineering" OR "software development" OR "software process" OR "SDLC" OR "DevOps" OR "continuous integration" OR "agile development")`

`ieee_exec2` `(("Monte Carlo simulation" OR "Markov chain" OR "stochastic process model" OR "process simulation") AND ("software" OR "process mining" OR "throughput" OR "lead time")) AND ("software engineering" OR "software development" OR "software process" OR "SDLC" OR "software testing")`

`ieee_exec3` `(("remaining time prediction" OR "lead time prediction" OR "cycle time prediction" OR "process forecasting" OR "outcome prediction" OR "event log construction") AND "software") AND ("software engineering" OR "software development" OR "software process")`

`ieee_exec4` `("mining software repositories" OR "software repository mining" OR "software archaeology" OR "mining commit logs" OR "GitHub mining" OR "Jira mining") AND ("process model" OR "workflow" OR "state transition" OR "Markov" OR "Monte Carlo" OR "stochastic" OR "process mining" OR "process analysis")`

The SpringerLink cell of Table 3 is the string that was run.

Appendix B. Included primary studies

This appendix summarises the 456 studies (publications; two conference–extension pairs are identified in Section 2.3). Table B.26 groups them by family. The identifier, year, role, and family pattern of each study are given

in Supplementary Material S1 (`supplement.pdf`) and in `included_studies_506.csv` of the replication package. The file `included_studies_340.csv` still contains the two papers from 1991 and 1992 (`dd71433d`, `37385fbd`) so that removal can be checked.

Table B.26: Classification of the 456 studies. TF-PM is process mining (IC4a), TF-ST a stochastic model (IC4b), TF-FC forecasting (IC4c), and TF-MSR repository mining (IC4d). The 306 remain from the coded list of 338 after 32 studies were removed because the title or abstract does not support inclusion. The 74 are the added studies read from an abstract or a PDF that meet at least one content criterion. The 76 title-only studies have no family mark.

Family pattern	**306**	**74**	**76**
TF-ST	51	44	–
TF-ST|TF-FC	81	6	–
TF-PM	70	9	–
title only, marks blank	–	–	76
TF-PM|TF-MSR	58	4	–
TF-ST|TF-MSR	15	1	–
TF-FC	6	4	–
TF-MSR	2	6	–
TF-PM|TF-ST	8	–	–
TF-FC|TF-MSR	5	–	–
TF-PM|TF-FC	3	–	–
TF-PM|TF-FC|TF-MSR	3	–	–
TF-ST|TF-FC|TF-MSR	2	–	–
TF-PM|TF-ST|TF-FC	1	–	–
TF-PM|TF-ST|TF-MSR	1	–	–
Total	**306**	**74**	**76**

Publications dated 2026 that were indexed by 12 April 2026 remain inside the eligibility window.

Appendix C. SEGRESS reporting checklist

Table C.27 maps each SEGRESS item to the location in this manuscript. Items marked N/A are those SEGRESS declares not required for mapping studies (no synthesis of primary-study outcomes).

Table C.27: SEGRESS reporting checklist for this systematic mapping study. Locations refer to the revised manuscript. Status: R = reported; P = partial; N/A = not required for mapping studies under SEGRESS.

Item	Requirement (mapping)	Where addressed	St.
1	Title identifies the report as a systematic mapping study and states the topic.	Title page: "Process Mining and Stochastic Modeling for Software Process Forecasting: A Systematic Mapping Study with a Human-Checked LLM-Assisted Screen". The subtitle names the report type and the title names the topic.	R
2	Structured abstract: background, objective, methods, results, limitations, conclusion.	Structured abstract (Context, Objective, Method, Results, Conclusion); the screen's Recall shortfall and its consequence for the absence claim are stated in Results and Conclusion.	R
3	Rationale: why the review is needed; gap vs prior secondary studies.	Introduction: states that no prior mapping covers the joint PM × stochastic × forecasting intersection.	R
4	Objectives / research questions.	Table 1; Section 2.1.	R
5	Eligibility criteria, date/language restrictions, handling of on-topic prior reviews.	Table 6 (IC1–IC3 publication filters; IC4a–IC4d content; EC1–EC5). Window 1 January 1994 to the search date, 12 April 2026. A 2026 publication counts only if it was indexed by that date. EC4 excludes off-topic secondary studies.	R
6	Information sources with search execution dates.	Tables 3 and 4; each export is dated 12 April 2026. All six Scopus strings (the six query names listed in Table 11), the four Web of Science strings, the two ACM strings, the four IEEE strings, and the SpringerLink string are reproduced in Appendix A (term lists in Section 2.2).	R

Continued on next page

Table C.27 — continued from previous page

Item	Requirement (mapping)	Where addressed	St.
7	Full search strategy (strings, snowballing, completeness checks).	Section 2.2. One snowballing iteration stores 595 records attributed to 400 source DOIs. The ten control papers (Table 5) are not those source DOIs. Recovering the ten known papers does not measure search recall.	R
8	Selection process: phases, number of assessors, disagreement handling, automation, agreement.	Sections 2.4, 6: one LLM screener; two human raters on the pre-registered sample (n = 468 at title/abstract, n = 177 at the second pass). Recall on the working-set sample is 0.630 (29 of 46). It is not a full-text Recall: a PDF was read for 0.5% of the 886 records. The round-1 FT row (Recall 0.246; κ = 0.122) is withdrawn. The two reviewers agreed with 92% of the suggested labels on the escalation set. The re-screen sheets (4,005 rows) match the suggestion on every row. The coded includes whose title or abstract does not support inclusion are the 32 in `labels_ta_insufficient_for_include.csv` (32 of 338, 9.5%), and they are removed.	R
9	Data-collection process: reviewers, independence, automation.	The same model extracts the fields. On 31 studies the exact match with a human rater on the free-text fields is 0%. Those fields were not replaced by a human re-extraction. There is no second human extractor for the full set; the human re-examination covers the 74 added studies and the 99 two-family studies.	R
10a	Outcomes for which data were sought.	N/A: mapping study does not synthesise primary-study outcomes.	N/A

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Table C.27 — continued from previous page

Item	Requirement (mapping)	Where addressed	St.
10b	Non-outcome variables and classification schemes.	Extraction schema: PM technique, stochastic technique, SDLC phase, dataset source, IC combination, reporting-quality scores (QA1–QA8).	R
11	Risk-of-bias methods (optional for mapping studies).	Reporting-quality scores (Dybå & Dingsøyr), not risk of bias. The same model assigns them. A score of 4/8 is not a filter on F1–F5, and no statement uses that cut.	R
12	Effect measures.	N/A (no outcome synthesis).	N/A
13 / 13a	Methods to analyse study characteristics; which studies enter each analysis.	Section 4.1 states the scope of every count once. Family-code counts use WS-151 (the 151 working-set studies that remain) and 293 joint-table studies; WS-151 is not the 150 added studies (ADD-150). The 169 and the 318 are the earlier cuts, before the removals named in Table 14. The corpus of this revision is 456 (Table 14). Quality scores do not decide which studies enter a count.	R
13b	Data preparation for outcome synthesis.	N/A (mapping).	N/A
13c	Methods for tables, graphs, maps of study characteristics.	Frequency tables, year plot, IC-combination counts, SPMF taxonomy, integration-level Table 15.	R
13d	Methods used to synthesise primary-study outcomes.	N/A (mapping).	N/A
13e	Sensitivity of the auxiliary corpus (not outcome synthesis).	Section 6, Table 21.	P
13f	Heterogeneity of primary-study outcomes.	N/A (mapping).	N/A
14	Reporting-bias / publication-bias methods.	N/A for mapping studies under SEGRESS.	N/A

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Table C.27 — continued from previous page

Item	Requirement (mapping)	Where addressed	St.
15	Certainty (GRADE) methods.	N/A for mapping studies.	N/A
16a	Selection results, ideally a PRISMA flow; agreement statistics.	Figure 1 ends at the corpus of 456: 340 distinct LLM confirmations, minus two pre-1994 papers, minus 32 coded studies whose title or abstract does not support inclusion, plus ADD-150, the 150 added studies after the eligibility pass (168 remained after de-duplication; 18 were then removed). Recall on the working-set sample is 0.630, and it is not a full-text Recall; auxiliary Recall is 0.725. The round-1 FT row (Recall 0.246; $\kappa = 0.122$) is withdrawn.	R
16b	Near-misses (optional for mapping).	EC3 is the modal FT exclusion; a full near-miss citation list is not included (optional).	P
17	Cite each included study and present characteristics.	The included-study list has 456 rows in `included_studies_506.csv` (the filename keeps the earlier count). The 340-row file is kept so the removal of the two pre-1994 papers can be checked.	R
18	Risk-of-bias results per study (optional for mapping).	Quality scores are descriptive, not a risk-of-bias judgement. The working-set scores are in Section 2.5. A score of 4/8 does not add or remove a study.	R
19	Results of individual studies (outcomes).	N/A (mapping).	N/A
20a	Analyses of study characteristics addressing each RQ.	Sections 4–5.1 (RQ1–RQ3, F1–F5).	R
20b–20d	Statistical outcome syntheses, sensitivity, heterogeneity.	N/A (mapping).	N/A
21	Reporting-bias results.	N/A (mapping).	N/A
22	Certainty of evidence.	N/A (mapping).	N/A

Continued on next page

Table C.27 — continued from previous page

Item	Requirement (mapping)	Where addressed	St.
23a	Interpretation in the context of other evidence.	Discussion / research agenda (Section 5.3).	R
23b	Limitations of the included evidence.	Optional for mapping; extraction-basis and Lost Evidence discussed in Section 6.	R
23c	Limitations of the review process.	Section 6. The mapping protocol was not pre-registered. The human check was. Two raters; Recall on the working-set sample is 0.630, below 0.90, and it is not a full-text Recall. The round-1 FT row (Recall 0.246; $\kappa = 0.122$) is withdrawn. F2 and F5 are withdrawn (free-text exact match 0% on 31 studies). F1, F3, and F4 keep the original family codes; the 99 coded studies with two technique families were re-examined by two of the authors for the third family (none moved to L2, none at L3; Section 4.1). The re-screen sheets (4,005 rows) match the suggestion on every row. The 32 coded includes whose title or abstract does not support inclusion (32 of 338, 9.5%) are removed.	R
23d	Implications for practice, policy, future research.	Section 5.3 (RA1–RA4 and the practitioner paragraph). Practitioner observations and the agenda are drawn from the cited studies; the counts F1–F5 situate but do not derive them (Section 4.1).	R
24a	Registration (or statement that the review was not registered).	The mapping protocol was not pre-registered. The human check of the screen was, before the round-2 judgements.	R
24b	Protocol availability (optional for mapping).	No standalone protocol deposit; methods are specified in Section 2 and the replication package.	P

Continued on next page

Table C.27 — continued from previous page

Item	Requirement (mapping)	Where addressed	St.
24c	Protocol amendments (optional for mapping).	Revised `maybe`/`pending` handling documented as a post-submission protocol change (Section 2.4).	R
25	Sources of financial / non-financial support.	Funding statement (CAPES Finance Code 001).	R
26	Competing interests.	Declaration of competing interest.	R
27	Public availability of data, code, and other materials.	Zenodo DOI 10.5281/zenodo.21939600. The included-study list has 456 rows in `included_studies_506.csv`. The 340-row file is kept for the pre-1994 removal check.	P

Source. Item definitions follow Kitchenham, Madeyski and Budgen, SEGRESS, *IEEE Trans. Softw. Eng.*, 49(3):1273–1298, 2023, doi:`10.1109/TSE.2022.3174092`.

CRediT authorship contribution statement

Rodrigo Almeida de Oliveira: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Juliano de Paulo Ribeiro:** Methodology, Validation, Investigation, Writing – review & editing. **Edson Emilio Scalabrin:** Conceptualization, Methodology, Supervision, Writing – review & editing, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Declaration of generative AI in the writing process

During the preparation of this work the authors used Claude (Anthropic) to review the manuscript text for language, internal consistency, and arithmetic. After using this tool, the authors (first author, advisor, and co-advisor) reviewed and edited the content as needed and take full responsibility for the content of the publication. Separately from the writing, large language models were used in the *conduct* of the study, as a screening and scoring instrument whose selection error the study measures; that use (models, prompts, stages, and the human check) is described in Sections 2.4 and 6.4, and the prompts, JSON schemas, and per-record decisions are deposited in the replication package (Zenodo DOI 10.5281/zenodo.21939600).

Data availability

Supplementary Material S1 (`supplement.pdf`) lists the identifier, year, role, and family pattern of each of the 456 included studies. A full replication package is openly available at Zenodo (DOI 10.5281/zenodo.21939600). The package contains: search strings for all five databases, deduplication and enrichment scripts, screening prompts and JSON schemas, the QA prompt, the cross-model κ verification scripts, all decision CSVs at every pipeline stage, the QA-scored 169-study and 212-study tables, the structured extraction tables (working-set and auxiliary), the inter-rater agreement reports (LLM–LLM and human–LLM), the 456-study list of this revision (`included_studies_506.csv`), the earlier 340-study file kept for traceability, a completed SEGRESS checklist, and a PRISMA flow snapshot.

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